

depends on the size of the valve and the specific application. Special control systems can also be used for automatic closing, e.g. see Table 1.

Fieldbus control

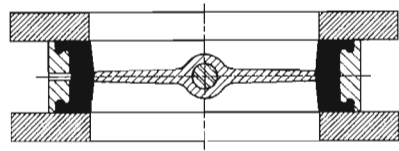
Butterfly valves have now become a viable alternative to conventional control valves due to development in sophisticated controls instrumentation and communications technology.

A major milestone in flow control is the introduction of fieldbus control systems.

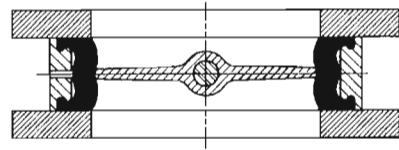
Fieldbus systems take advantage of developments in low-cost microchip technology, allowing intelligence to be built into each device in the loop. Each device communicates via a protocol developed for rapid control information, with the ability to adjust a valve's position and enable the speed of the valve stroke to be altered using programmable, menu-driven communications devices.



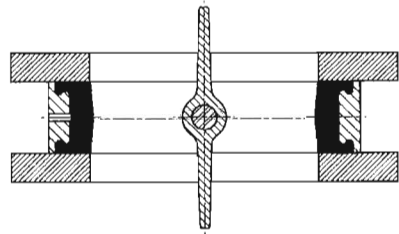
Figure 8.



Closed



Closed, inflated



Open, deflated

Figure 9.

The diagnostic role of a fieldbus system means it can give advance warning of potential problem areas so that corrective action can be taken. This instrumentation technology has resulted in a new breed of valve controls instrumentation.

See also the chapter on Valve Actuators.

Inherent flow characteristic

The inherent flow characteristic of a valve has been defined so as to keep the differential pressure across the valve constant. When the differential pressure across the valve is constant, the flow rate (q) through the valve is proportional to the valve flow coefficient (C_v). Because the valve flow coefficient (C_v) reflects the effective flow cross-section of the valve, the valve inherent flow characteristic

Table 1. Selection table for automatic-closing butterfly valves

	Is power available? ... During emergency closing? ... Any voluntary remote control?	No		Yes				
	Electrically motorised valve							•
Control device	Hydraulic jack	•	•	•	•	•	•	
	Counterweight	•	•	•	•			
	Oleopneumatic accumulator					•		
	Mechanical locking							
	Manual resetting	•	•					
	Automatic resetting			•				
	Automatic leakage recovery				•	•		
	Motorised pump		•	•	•	•	•	
	Hand pump	•						
Control signal	Defect transmission							
	Mechanical	•						
	Electrical						•	•
	Optional		•	•	•	•		
	Electrical remote control			•	•	•	•	•

shows how the effective flow cross-section changes as a function of the relative travel (h).

Figure 10 shows the most common valve inherent flow characteristics as a function of the relative flow coefficient (N) and the relative travel (h).

Valve sizing

The size of butterfly valves to be used for control purposes should not be dictated by the nominal diameter of the pipe, but should be calculated on the basis of the operating characteristics in order to achieve the correct control characteristics.

To determine the size of a control valve, the opening angle characteristics need to be considered. Typically, some butterfly valves are designed with approximately equal percentage characteristics over an opening of 60° .

The example in Figure 11 shows how to calculate the K_v (flow coefficient) from liquids and gases.

Pressure/temperature ratings

The maximum working capability of a valve is either the body rating or the seat shut-off capability, whichever is the lower.

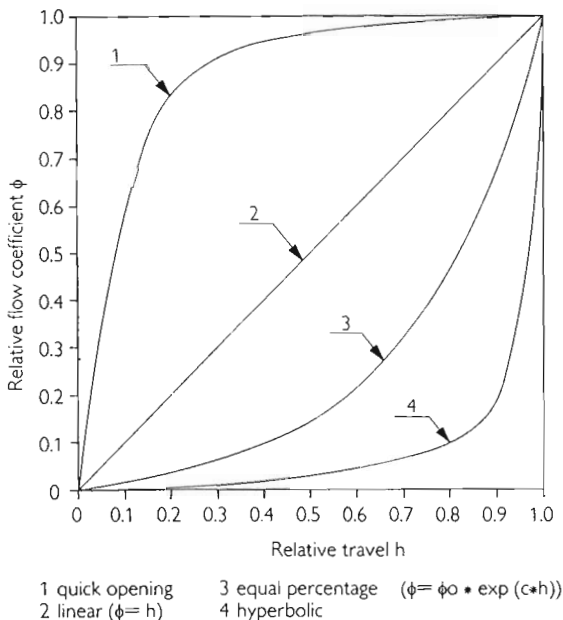


Figure 10. Valve inherent flow characteristics.

METHOD

Calculate the Kv (flow coefficient) from:

Where

a) Liquids

$$K_v = Q \sqrt{\frac{\gamma}{\Delta p}}$$

Liquid sizing formula is only applicable for subcritical flow

b) Gases

$$K_v = \frac{Vn}{514} \sqrt{\frac{G.T}{\Delta p.P_2}}$$

Gas formula only applicable if the ratio Δp to P_1 (upstream pressure absolute) is less than 0.10

K_v = Flow coefficient (Metric Units)
 C_v = Flow coefficient (Imperial Units)

Q = Max. Flow volume in m³/h

γ = Specific weight in Kg/dm³

F = Cross section of pipe in cm²

Δp = Pressure drop in bar

Vn = Max flow volume in Nm³/h

G = Specific weight in Kg/Nm³

T = Absolute Temperature in °K

P_1 = Absolute pressure upstream in bar

P_2 = Absolute pressure downstream in bar

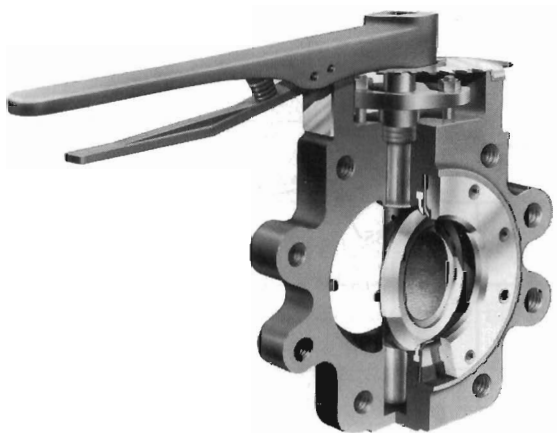
Having calculated the K_v , the figure obtained should be compared with these below and the normal diameter determined.

DN	Cross-section of pipe in cm ²	Opening Angle																	
		90° Cv	Kv	80° Cv	Kv	75° Cv	Kv	70° Cv	Kv	60° Cv	Kv	50° Cv	Kv	40° Cv	Kv	30° Cv	25° Cv		
50	19.6	111	130	89	104	76	89	59	68	45	53	23	27	14	16	7	8	5	6
65	33.2	170	198	136	159	111	130	89	104	70	82	35	41	22	26	12	14	8	9
80	50.3	256	299	205	239	175	204	136	159	106	124	53	62	32	37	18	21	12	14
100	78.5	420	548	405	473	341	398	260	303	200	233	102	119	62	72	35	41	23	27
125	123	661	854	627	734	534	627	418	488	213	249	132	154	75	88	51	60	35	41
150	177	1066	1444	1066	1244	880	1027	683	797	350	408	213	249	123	144	83	97	55	64
200	314	2777	3740	2777	3240	1880	2194	1405	1744	1113	1296	599	699	359	418	213	249	145	169
250	491	4273	5906	4273	5006	3076	3589	2350	2742	1837	2144	982	1146	572	667	333	389	222	259
300	707	6410	8840	6410	8033	4273	5006	3463	4039	2649	3091	1367	1595	854	996	470	548	324	378
350	962	8547	9973	8223	8078	5726	6687	4258	5095	2880	2944	1880	2194	1112	1296	641	748	427	498
400	1257	10683	12446	9230	10770	7692	8975	5555	6492	4358	5085	2264	2642	1452	1694	769	897	555	648
450	1590	14352	17433	11985	13961	10256	11967	7863	9175	6068	7081	3162	3690	1965	2293	1068	1246	769	897
500	1963	18803	21940	14957	17452	12820	14959	9829	11469	7435	8676	3931	4587	2393	2792	1367	1595	961	1121
600	2827	28831	34924	20512	28955	17521	20445	14102	16455	10042	11645	5213	6083	3247	3789	1880	2194	1282	1496

Figure 11. Example method for calculating flow coefficients K_v/C_v .



Ceramic-lined butterfly valves have provided a long-term, low-maintenance solution to controlling abrasive and acidic titanium dioxide slurry flows.



High-performance butterfly valve.

In Figure 12 the seat ratings shown are based on data from API 609 and the body ratings shown are from ANSI B16.5/BS 1560 part 2.

Butterfly check valves

Butterfly check valves differ from conventional butterfly valves in employing a hinged instead of a pivoted disc, with a sealing ring around the edge of the disc (Figure 13). With forward flow, the two halves of the disc are swung together to trail downstream. With reverse flow, the two halves open to approximately 45° each, sealing the bore.

With this mode of working, a butterfly check valve is, in fact, another form of flap valve. It has the advantage of rapid action with a resilient seal.

Butterfly check valves (Figure 14) require only a short length of body which can virtually be a length of standard pipe flanged at both ends (or threaded or plain ends in smaller sizes). It is also produced in wafer form for clamping between two pipe flanges. In this case the body length only needs to be sufficient to accommodate the hinge post and the two valve plates in their closed position.

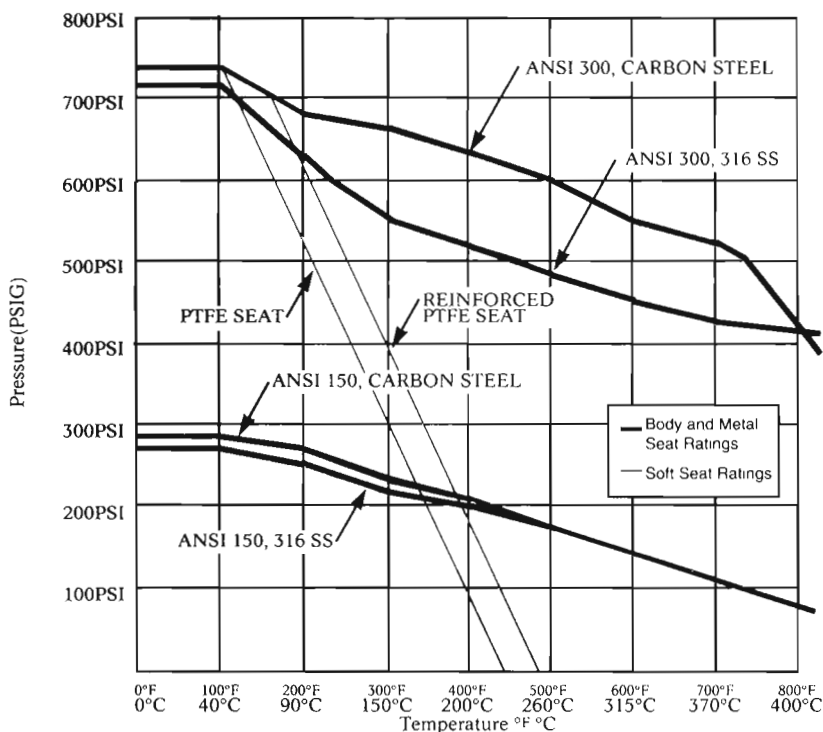


Figure 12. Typical pressure/temperature ratings.

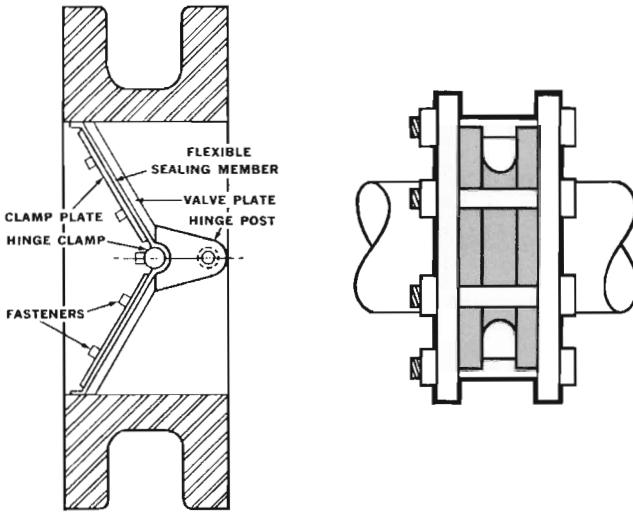


Figure 13. Butterfly check valve.

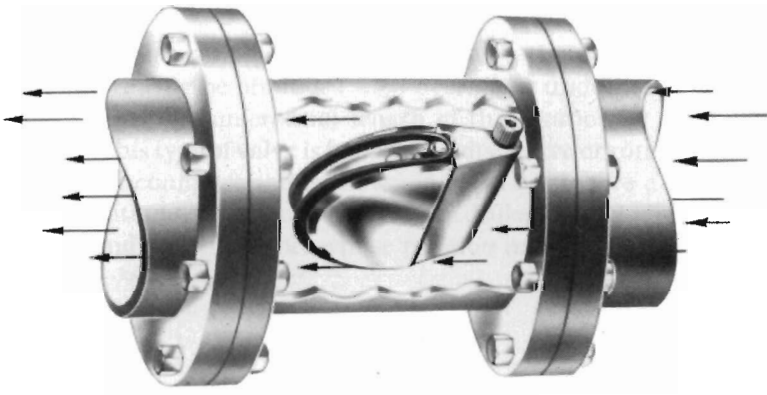


Figure 14. Butterfly check valve.

Advantages of butterfly valves

- Butterfly valves are compact, require less space, less weight, less raw material and are generally less costly than other valve types.
- Butterfly valves are quarter-turn valves, easy to operate by lever, gearbox or actuator. They have excellent gland integrity.
- Butterfly valves have few parts and are generally easy to maintain.
- Butterfly valves are suitable for control purposes with linear flow characteristics between 30° and 70° of opening.
- Butterfly valves offer excellent rangeability from 30:1 to 100:1.

Typical major areas of application for butterfly valves include: liquids, gases, steam, cryogenics, pulpstocks both on control and shut-off services in chemical processing, waterworks duties, oil and gas processing, pulp and paper manufacture, the power industry, refineries and food and beverage processing.

Properly selected and integrated with the latest controls and communications technology, butterfly valves can provide an economical, long-term solution to flow control in large process plants.

Rotary Disc/Rotor Valves

The rotary valve is an obvious configuration, although not widely used. It can be classified as a type of spool valve where the ports run axially instead of circumferentially (Figure 1). Sealing in the closed position, valves are a close fit over a restricted area only. Also friction can be high. In these respects, it is inferior to a conventional spool valve or even a plug valve.

A somewhat better configuration is to employ a 'solid spool' with drilled-through ports (Figure 2); or the rotary plate valve (see the chapter on Spool Valves).

Geometric design is simplified to a degree with these latter types of valves, and both can readily be produced with overlap or underlap as required, e.g. by enlarging the circumferential length of the rotation or stationary port, respectively. This type of valve is known as a disc valve or rotary disc valve.

A third basic configuration is shown in Figure 3, where a spindle or shaft rotates in a stationary housing. The shaft is drilled axially and also vertically at the blind end to form a port whose position matches the inlet port in the

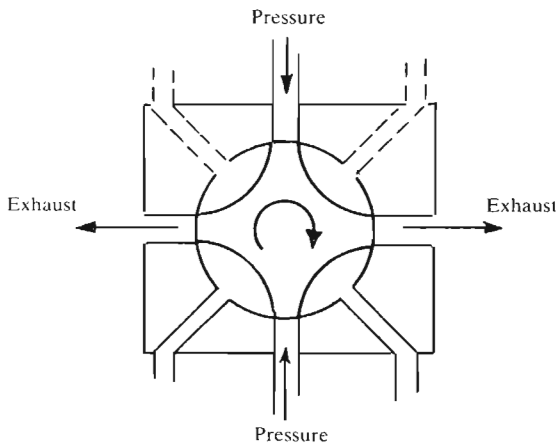
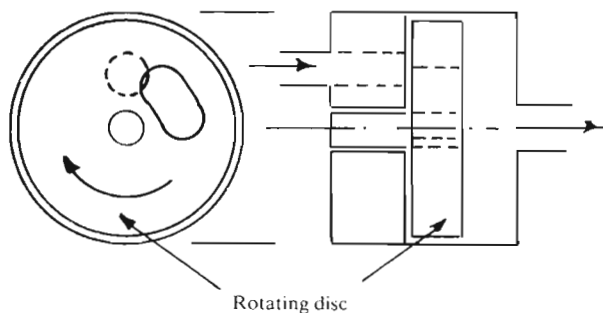
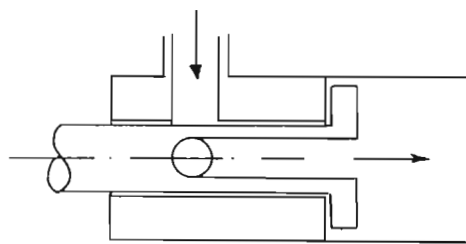


Figure 1.

*Figure 2.**Figure 3.*

stationary member. Again, the port opening(s) can be modified in length to provide overlap or underlap as required.

A feature of both rotary valve types shown in Figures 2 and 3 is that they are capable of providing exact timing of port opening with continuous rotation of the rotary element.

The design of rotary valve shown in Figure 4 overcomes the problem of friction and leakage by using shear seal rings in each port and a thrust bearing between rotor and housing. The shear seal rings are grooved to retain O-ring seals and are spring-loaded onto the smooth rotor face by individual disc springs. The flow passages through the valve are made as clear as possible to reduce the pressure drop through the valve.

Rotor valves

This type of valve differs from the typical rotary valve in that the valve design allows for transitional flow to occur when changing valve positions. Transitional flow eliminates dead-heading problems associated with positive displacement pumps.

Typically the rotor valve (Figure 5) consists of four flow types with many flow-pattern combinations and 'piggable' versions. The stem and rotor are

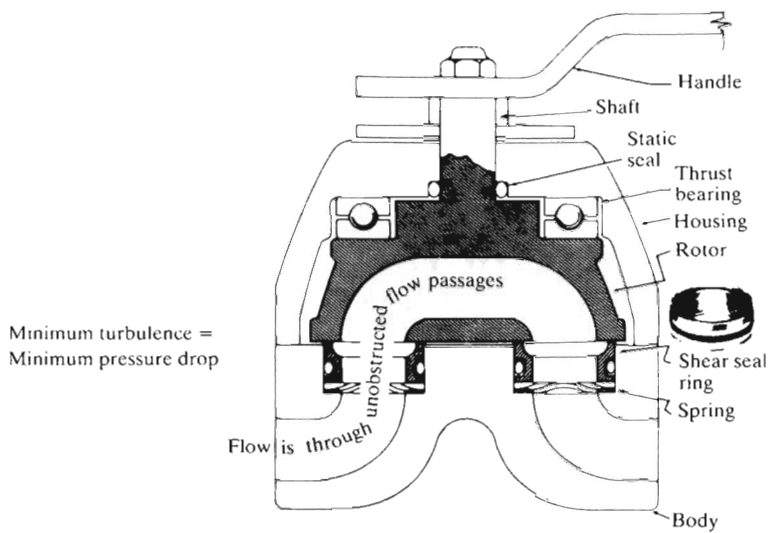


Figure 4.

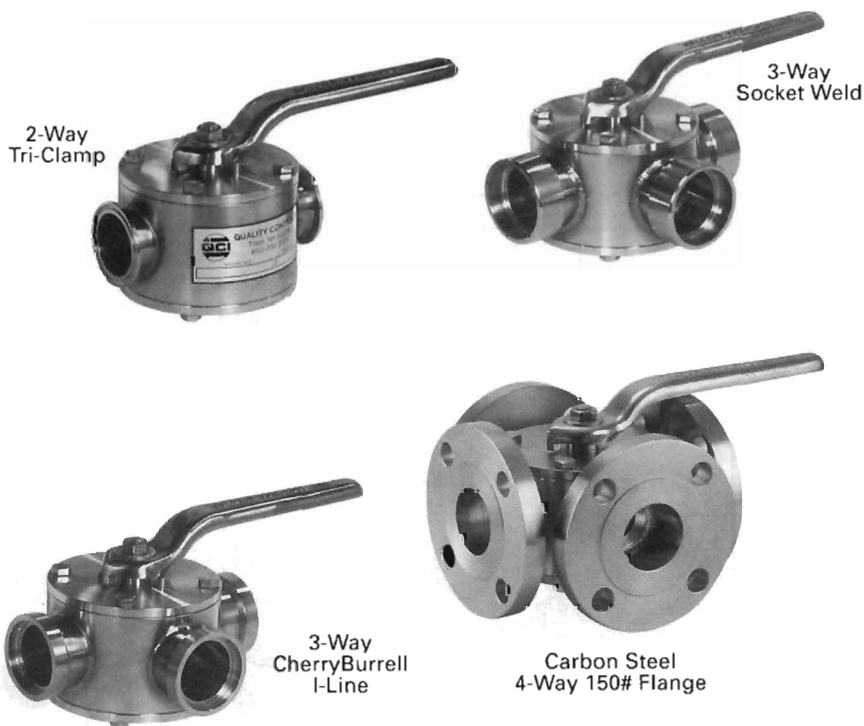


Figure 5. Examples of rotor valves.

usually of a one-piece construction that can eliminate the source of wear and repair common to the typical ball valve design. Full-port diameter reduces the pressure drop across the valve, thereby increasing the flow. An independent leaf seal design eliminates the large cavities common to the ball valve and also the need for cavity fillers, which still create stagnant seams for product to enter.

VALVE FLOW PATTERN COMBINATIONS

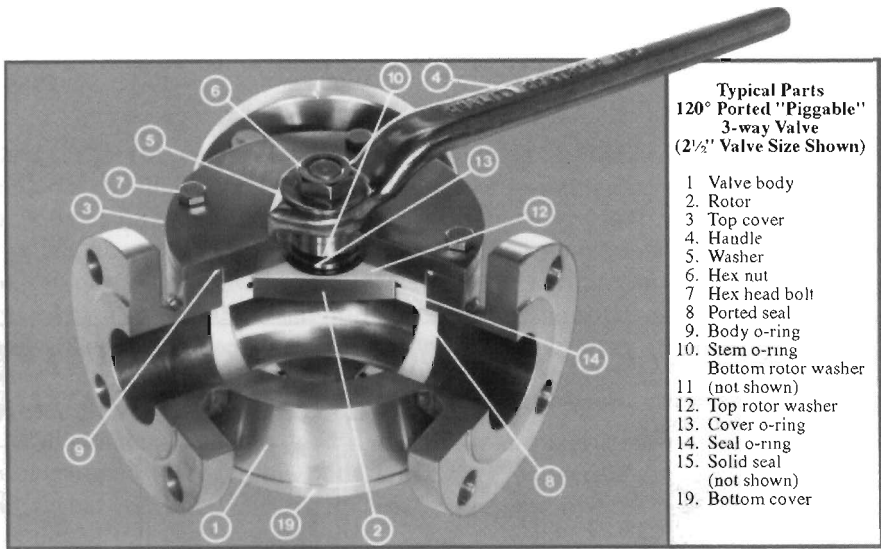
PORTING	2-WAY	3-WAY		4-WAY			
	S-	L-	T-	LL	**S-	**L-	**T-
Position 1							
Position 2							
Position 3							
Position 4							

PORTING	5-WAY						
	B-	BS	BL	BT	B- (with special porting)		
					A,B,C&E	A,C&E	A,B&E
Position 1							
Position 2							
Position 3							
Position 4							

Figure 6. Valve flow-pattern combinations.

Rotor valves of this type are favoured by a number of industries including: food and beverage, pharmaceutical, chemical, petrochemical, refining, paper and paint and particularly where quick-couple fittings for sanitary systems are preferred.

Typical flow-pattern combinations are shown in Figure 6.



120° ported piggable 3-way valve.



Figure 7. De-mountable rotary valve.

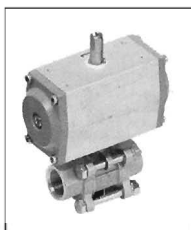
De-mountable rotary valves (Figure 7) are mainly used for sanitary duties in dairy, food, pharmaceutical and chemical applications with powder or granular product. This type of valve is usually produced in 316 stainless steel with the rotor end-plate and shaft seals easily disassembled without tools and the unit capable of being cleaned in place (CIP). Capacities up to 1500 cubic ft/hr in either drop-through or convey-through configurations are common.



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Globe Valves

Over the last 60 years, globe valves have been re-appraised in the light of new materials available for use. They are popular as industrial valves because they have fine adjustment and permit unobstructed opening through pipelines.

The globe valve is characterised by a baffle or partition separating the two halves of the body, with an interconnecting part at the centre opened and closed by a screw-down/screw-up disc or plug mounted at right angles to the body (Figure 1). The name derives from the fact that original body shapes were spherical, the more usual modern form being semi-spherical or even substantially parallel-sided.

The globe valve offers good regulation characteristics but high resistance because of the tortuous flow path. This can be reduced to some extent by making the throat area equivalent to that of the pipe (calling for a more bulbous body),

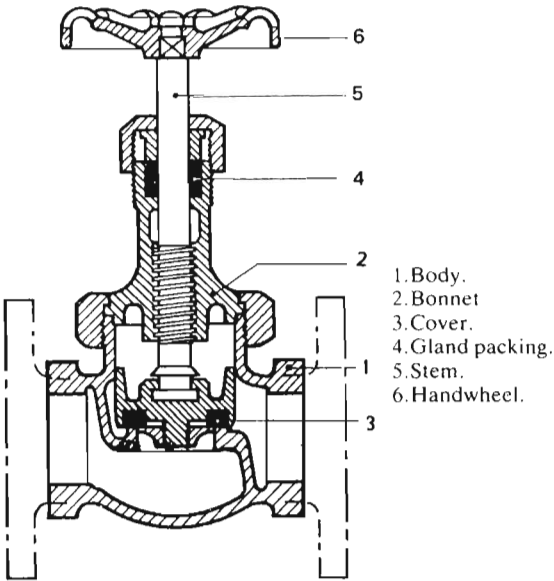


Figure 1. Screw-down globe valve.



Bronze globe valve.

rounding the partition to smooth flow or inclining it to the flow (Figure 2). Reduction of head loss by such treatment is generally minimal and so right-angled partitions are commonly used.

Globe valves are produced with a variety of discs or plugs and seals. Discs provide line contact with the seat which can be broken by solid deposits forming on the seat. They are thus mainly suited only for clean fluids. Disc valves, too, are not as effective as plugs for throttling duties and so are normally used only on shut-off valves. Another limitation of the disc is that it does not provide positive shut-off for air and gases, unless made of a resilient material.

Plugs are used in a variety of configurations, ranging from needle shapes to semi-discs. The plug contour also governs the throttling characteristics, e.g. equal-percentage plugs (percentage flow proportional to valve lift), etc. Needle plugs provide the finest flow control. Globe valves fitted with this latter type of plug are normally referred to as needle valves. Needle valves are normally made only in small sizes.

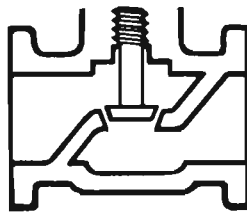
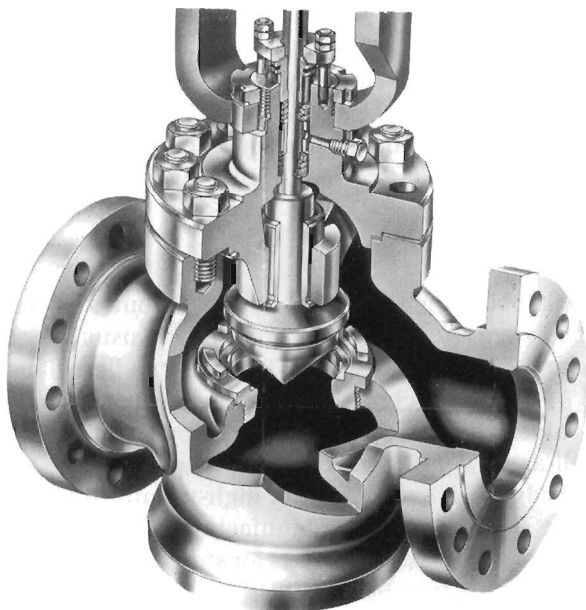
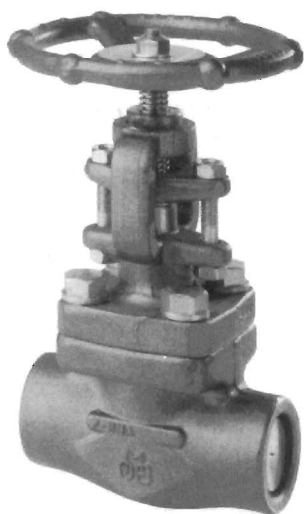


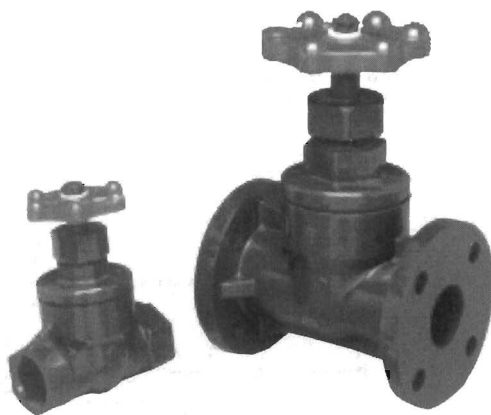
Figure 2.



Cutaway of globe valve.



Forged-steel globe valve.



Plastic globe valves.

Seats

Globe valve seats may be cast integral with the body or take the form of screwed-in, pressed-in, or spot-welded rings. A variety of materials may be used for seats, depending on the application, including coated seat rings with plastic inserts. Normally only screwed-in seat rings are replaceable.

Stems

Possible stem assemblies include inside screw and outside screw (both rising stem) and sliding stem. Inside stem rising screw is usual. A stem seal (stem gland) is necessary to eliminate leakage. This is normally a gland-type packing or gland rings. A diaphragm bonnet seal or bellows bonnet seal may also be used on globe valves, the former isolating the working parts of the valve from the fluid as well as preventing leakage to the atmosphere. Metallic bellows seals are often used on valves intended for high-vacuum duties.

The valve shown in Figure 3 is manufactured from malleable iron. This valve is widely used on standard pipelines for steam, oil and water.

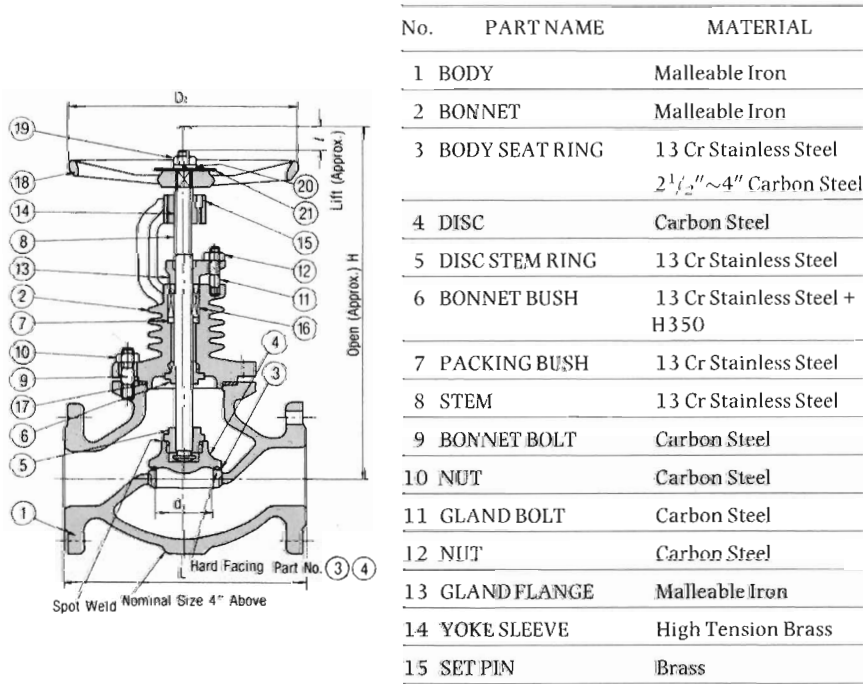


Figure 3. Malleable-iron globe valve.



'Y'-pattern globe valve: bellows seal, inside screw and yoke.

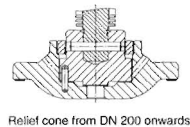
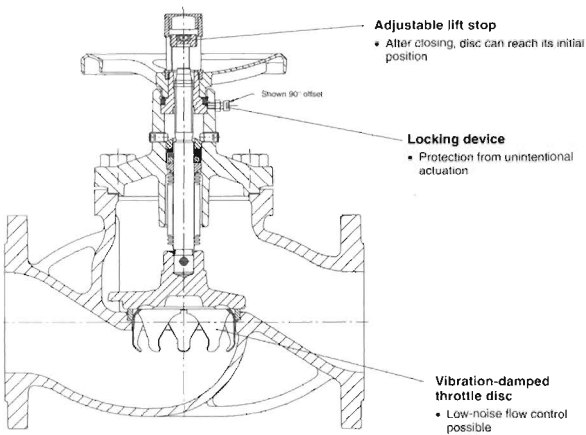


Figure 4. Globe valve with metallic seat.

Malleable iron provides a high yield point, a more important property than tensile strength for valve materials. It is also stable against temperature change. Low-pressure valves of this type generally have PTFE seats. Valves for use at higher pressures have the valve disc and body seat ring made from hard-faced stainless steel. The hard face is an alloy containing cobalt, chromium and tungsten. Valves of this type can also be fitted with replaceable PTFE discs when used with liquefied petroleum gases. The gland packings are usually of a synthetic rubber or butadiene-acryl-nitrile polymer V packing design to provide for good oil and wear resistance. Over time, the V packing will harden; tell-tale signs are: a friction sound when the valve is opened and

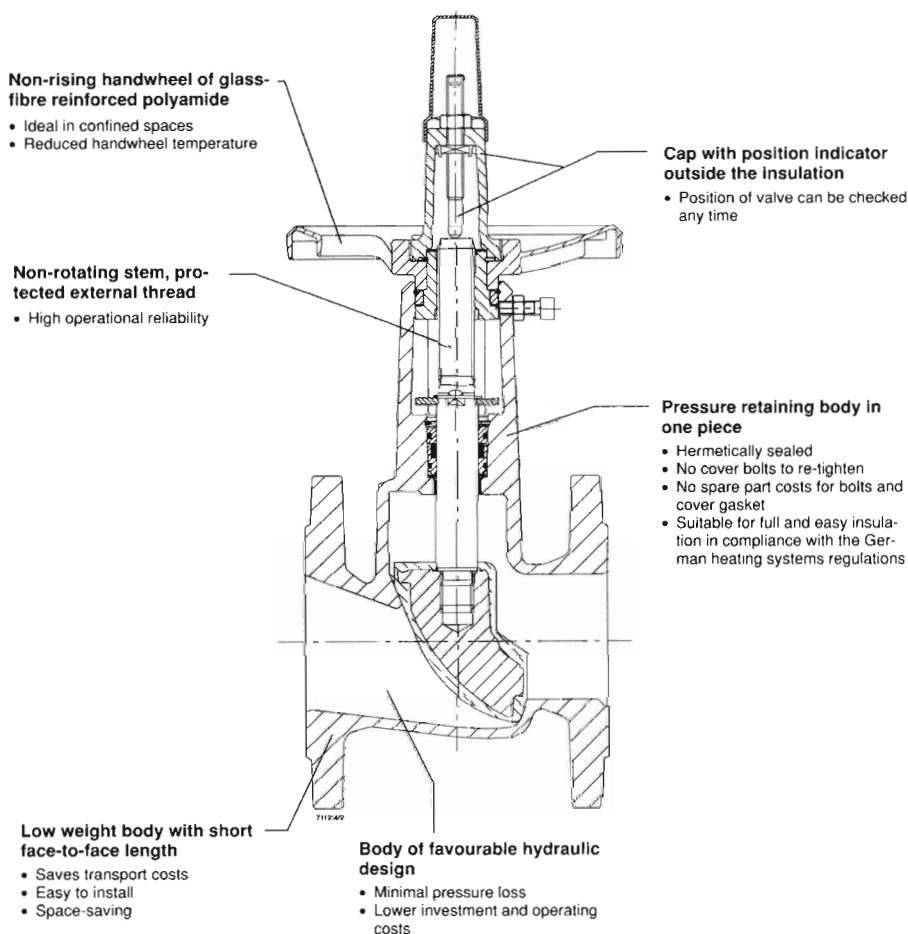


Figure 5. *Globe valve with high-temperature soft seat.*

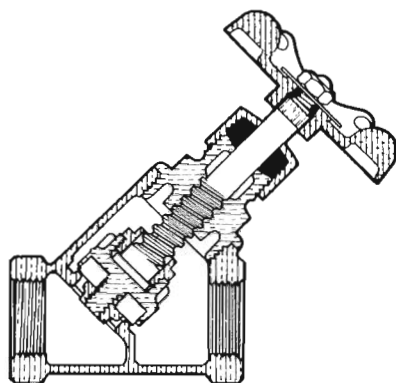


Figure 6. Oblique or 'Y' valve.

closed; the opening and closing operation becomes difficult to operate; and the gland actually leaking. The packing must then be replaced.

An example of a standard globe valve for use in hot- and warm-water heating and boiler installations is shown in Figure 4. This particular valve has a metallic seat. The temperature range is from -10°C up to $+350^{\circ}\text{C}$, with pressures up to $p = 25$ bar.

The valve illustrated in Figure 5 is fitted with a soft seat for high-temperature operation from -10°C to $+200^{\circ}\text{C}$. Typical applications included hot-water heating systems, district heating plants and low-pressure steam plants. The valve has a short face-to-face length which makes it particularly compact. Globe valves are a traditional standard solution for many control applications because of the ability to modify the trim design for different throttling purposes.

Oblique valves

The oblique valve or Y-valve is a hybrid globe valve characterised by the stem being angled (Figure 6). As a consequence the flow path is less tortuous, with reduced pressure drop compared with a conventional globe valve. It retains the same good throttling characteristics as a globe valve and can be fitted with similar types of plugs.

Construction is basically the same as globe valves with the option of integral or fitted seat rings, stem options and stem seal gland treatment.

Gate Valves

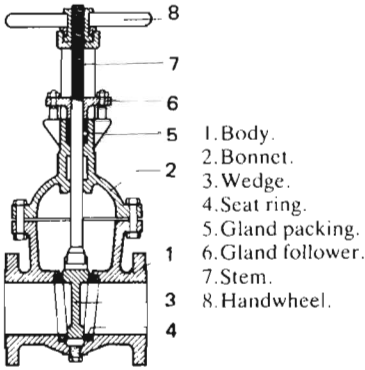
The main feature that distinguishes a gate valve is the flat face or vertical disc or wedge that slides in a track or seat which can be lifted in a direction at right angles to the valve until clear of the flow path. Generally gate valves are used for on–off non throttling service, i.e. they are intended to be either fully open, when they offer little resistance to flow, or fully closed. For this reason, they are the principal valves used in bulk pumping practice.

Large gate valves tend to be power operated. Throughout the water industry, for example, automation with actuated valves has been introduced in response to the requirement for lower manning levels.

Gate valves are divided into a number of classes, depending on the design of the ‘gate’ and its seating faces.

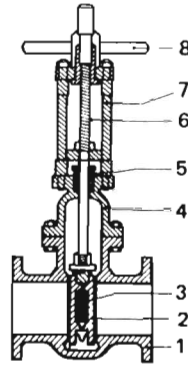


Gate valves.



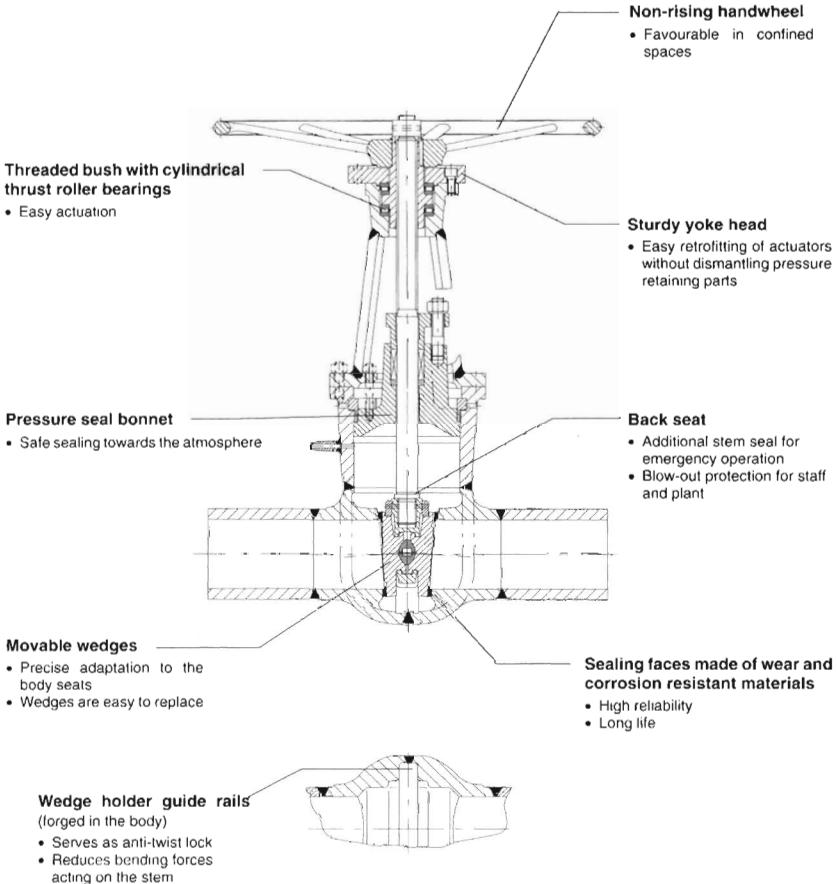
1. Body.
2. Bonnet.
3. Wedge.
4. Seat ring.
5. Gland packing.
6. Gland follower.
7. Stem.
8. Handwheel.

Solid wedge gate valve.



1. Body.
2. Disc.
3. Spring.
4. Bonnet.
5. Gland.
6. Stem.
7. Yoke.
8. Handwheel.

Parallel slide gate valve.



Gate valve with pressure seal bonnet.

Wedge-gate valves

The gate is wedge-shaped and seals on corresponding faces in the faces of the valve body. Wedges and seats can be made of, or coated with, resistant material or faced with plastic such as PTFE. The plastic is often contained in a groove to prevent it spreading. Figure 1 shows a wedge-gate valve with external screw, and Figure 2 one with internal screw.

Flexible-wedge valves and split-wedge valves are similar to the aforementioned, but with provision for slight seat misalignment.

Wedge-gate valves can be further described as inside-screw or outside-screw patterns (see Figure 3). They are widely used for oil, gas and air services, and also for handling slurries, etc.

Double-disc valves

In these valves the gate is in the form of two discs which are forced apart against parallel seats by a spring. This provides tight sealing without relying on fluid pressure, making this type of valve particularly suitable for steam duties as well as handling gases and light oils.

Parallel-slide valves

In 1886, Joseph Hopkinson introduced the parallel-slide valve where sealing of the valve relies on the upstream pressure acting on a flat parallel gate valve.

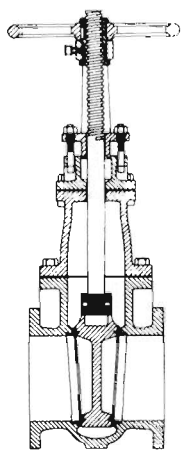


Figure 1. External screw.

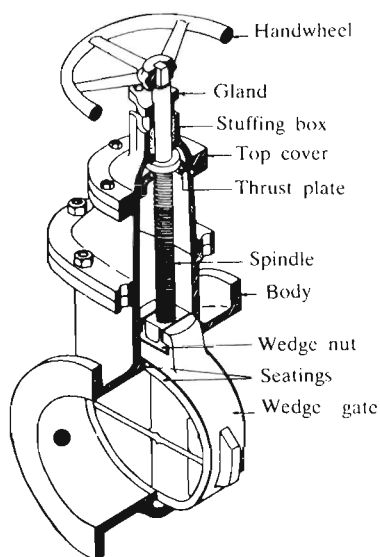


Figure 2. Internal screw.

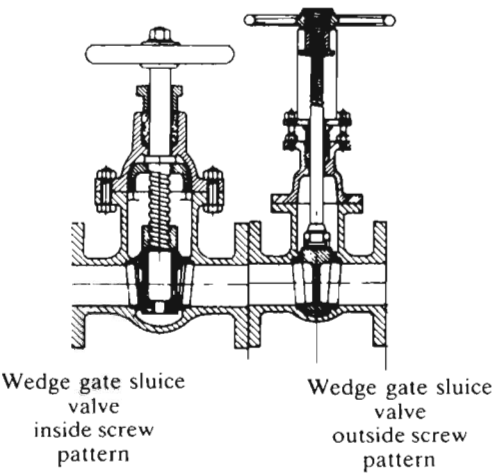
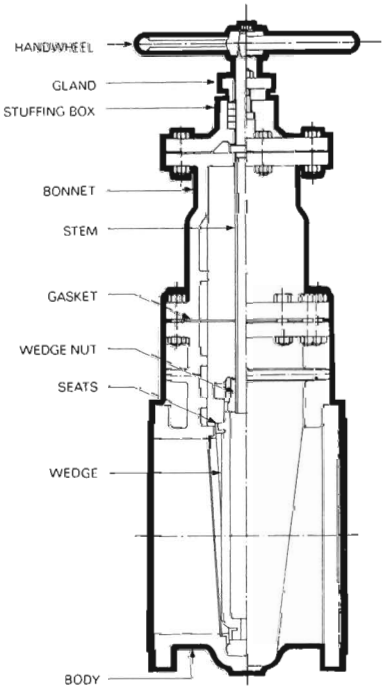
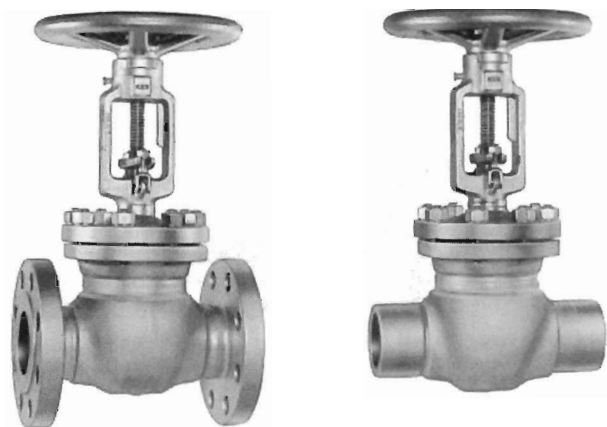


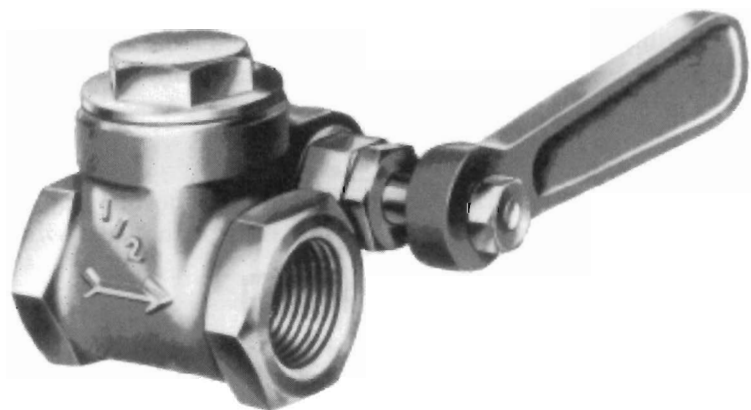
Figure 3.



Inside-screw gate valve.



Flanged and butt-welded gate valves.



Bronze double-disc lever gate valve.

The system is still popular today and the simplest (and probably most effective) layout employs two discs as valve members initially separated by a spring (see Figure 4). The function of the spring is to prevent the discs from rattling and to encourage a wiping action on the downstream disc when under pressure and on both discs when there is virtually no pressure in the line.

This is to avoid—as far as is possible—grit or scale becoming trapped between the vulnerable seating faces which might impair their sealing properties.

Parallel-slide valves are used extensively in the pulp and paper industry.

Sluice valves

This is a name applied to solid-wedge valves for waterworks.

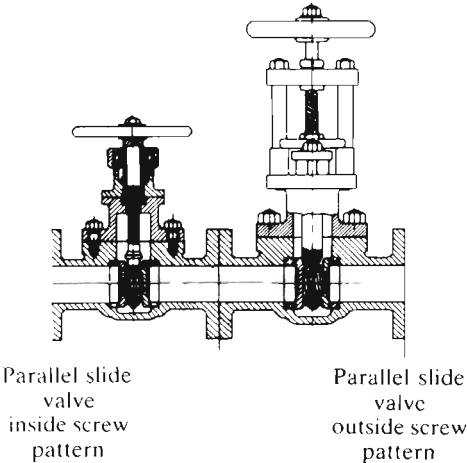
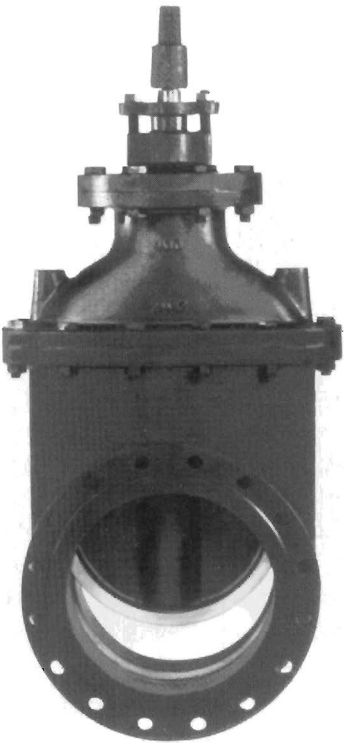


Figure 4.



Metal-seated gate valve.

Bellows seal-gate valves

Environmental protection is a major concern and the bellows seal valve is designed to minimise exposure to dangerous or harmful substances through valve-stem leakage.

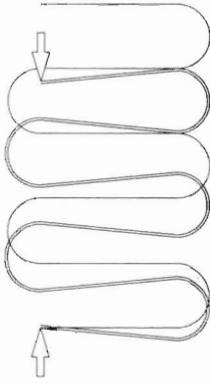


Figure 5.

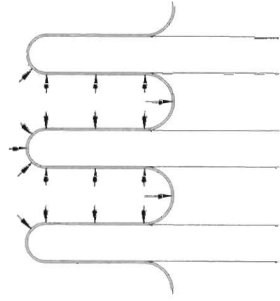


Figure 6.

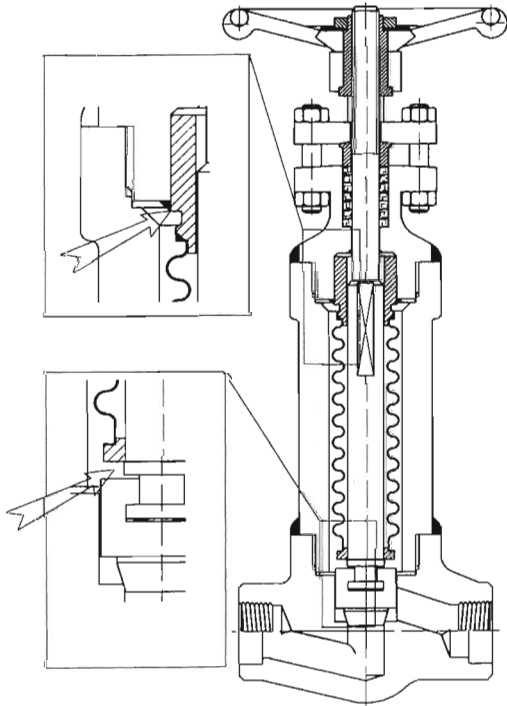
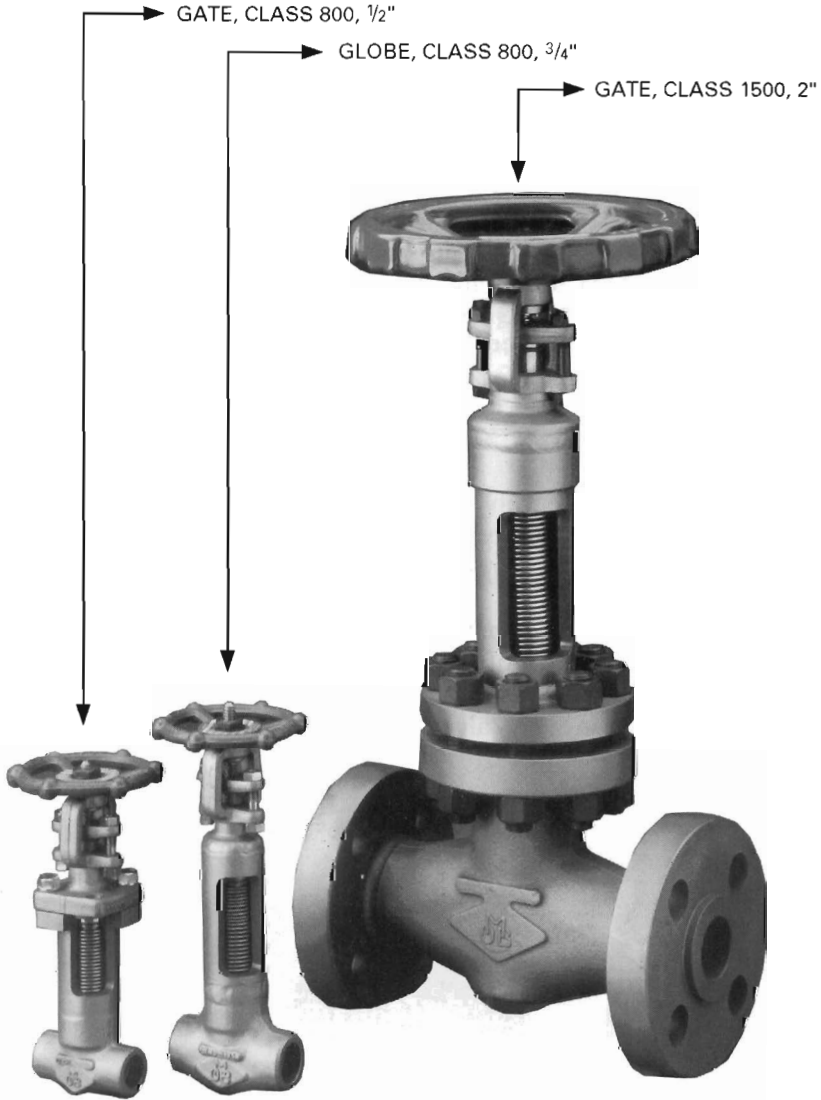


Figure 7. Construction details.

The bellows is a metallic device capable of sealing between the valve stem and the bonnet to prevent escape of the system fluid to the atmosphere. The bellows take the form of convolutions that can move linearly. A hermetic seal is achieved by welding the bellows to the valve stem at one end and to the

BELLOWS SEAL VALVES



Bellows seal valves.

bonnet at the other end (Figure 5). The structural shape of the bellows provides resistance to high pressure, even with thin wall thicknesses (Figure 6).

In operation, the bellows eliminates a leak path to atmosphere. The stem/packing area is sealed from the medium being processed and the bellows becomes the primary seal since the bellows assembly is welded to the stem and to the bonnet as shown in Figure 7.

Replaceable bellows are gaining popularity with valve users in process plants.

In replaceable bellows valves, the bellows is not welded to the bonnet; instead, it is welded to a transition piece that is clamped between the body with standard gaskets to seal the joint. The lower end of the bellows assembly is welded to the disc which is attached to the stem by a threaded connection. The gasket on the media side is generally a spiral-wound gasket. The gasket at the top of the bellows assembly only comes into operation if the bellows fails.

Where maintenance is difficult, the welded bonnet valve may be the better choice.

Gate valves are used mainly in general industry, power stations, process engineering, pulp and paper and marine engineering for water, steam, gas, oil and non-aggressive media.

Actuation

Manual actuation of gate valves is invariably by screw and handwheel. The screw mechanism may be exposed or protected and the screw 'rising' or 'non-rising'. A variety of materials for the working parts is offered by some makers.

Power actuators are very often fitted, especially where valves are difficult to access and are operated frequently. Automation and semi-automation control schemes also make extensive use of actuators.

See also the chapter on Slide Valves.

Needle Valves

Small sizes of globe valves fitted with a finely tapered plug are known as needle valves. This description also applies to any type of valve incorporating a tapered needle having axial movement relative to the axis of a concentric orifice and thus controlling the effective opening of the orifice.

Three basic configurations are shown in Figure 1: (A) is a simple screw-down valve; (B) is an oblique version, offering a more direct flow path; (C) is another form where the controlled outlet flow is at right angles to the main flow (and may be distributed through one or more passages). In these basic

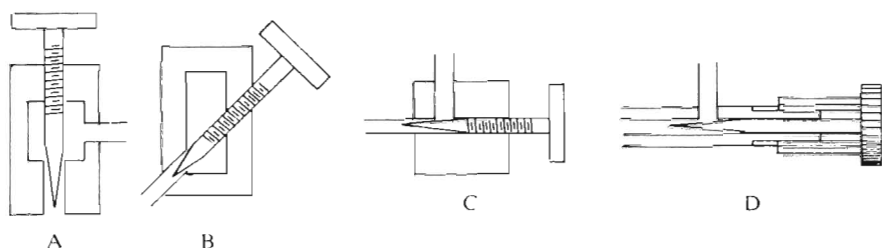
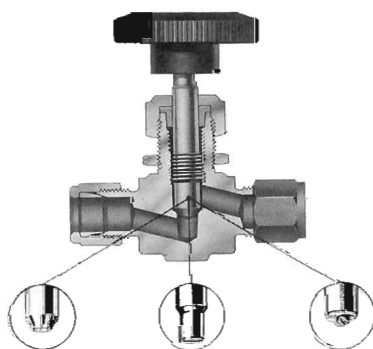
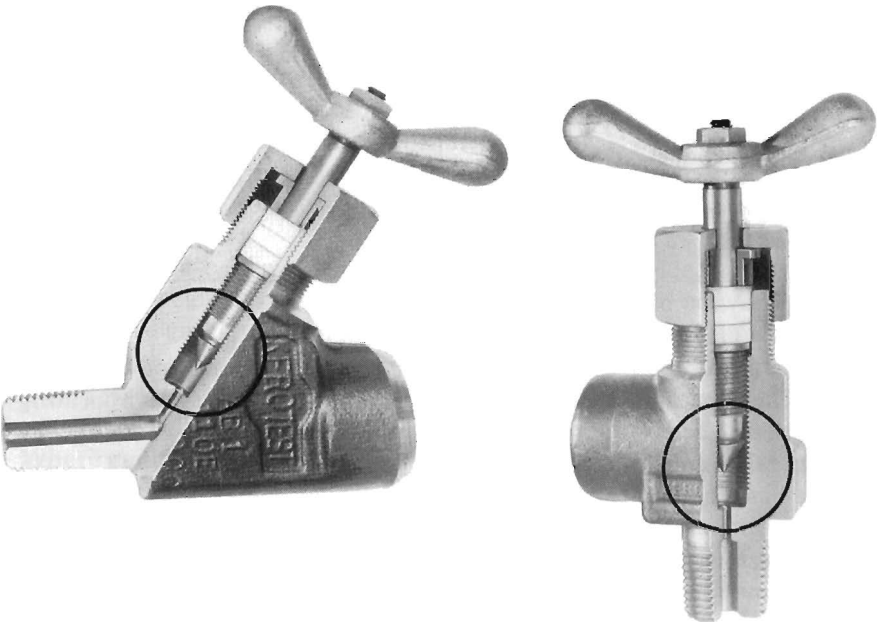


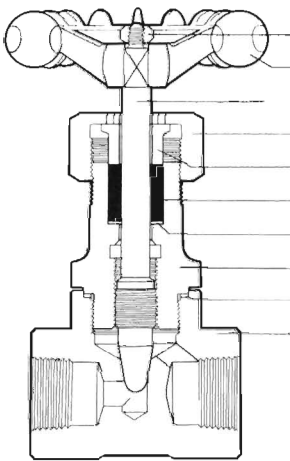
Figure 1.



Needle valve with interchangeable stem.



Oblique and 'T'-type needle valves typically used for process duties.



Part	Material	B S	A S T M
Handwheel Nut	Carbon Steel	1768	
Handwheel	Malleable Iron	309	
Stem	Stainless Steel	1506 713	A182 GR F6
Gland Nut	Carbon Steel	1506 111	
Gland	Carbon Steel	1506-111	
Packing	Valves and Plastic Non-Metallic		
Steel Packing Ring	Mild Steel	970 EN1A	
Bonnet	Carbon Steel	1506 111	
Bonnet Gasket	Armco Iron		
Body	Carbon Steel Forging	1503 161 GR 32	A105 GR 2

Typical needle-type meter valve.

versions a threaded needle is shown, the thread itself acting as a seal to eliminate leakage past the needle. This is normally quite satisfactory in very small sizes of needle, although a more leak-tight arrangement is to mount the needle in an externally threaded end piece [Figure 1 (D)]. This end piece can also act as a grip for adjustment of the needle. Sealing in this case can be further improved if necessary by incorporating an internal seal, such as an O-ring. The other common form of needle valve is the float-controlled, carburettor-type valve.

See also the chapter on Globe Valves.

Pinch Valves

The ‘working’ element of a pinch valve, also known as a clamp valve, is an elastomeric tube or sleeve which can be squeezed at its mid-section by some mechanical system until ultimately the tube walls are pinched or clamped together producing full closure of the flow path.

In its simplest form it can consist merely of a length of elastomeric tube fitted with a pinch bar mechanism incorporating a closure stop to prevent over-pinching of the tube. More usually the moulded rubber tube is housed in a metal body which also incorporates the pinching mechanism (Figure 1). This can be a simple screw-operated mechanism, where the pinch is applied only to one side of the tube, or a differential screw controlling two pinching mechanisms working in vertical opposition. The latter produces lower-stress working of the tube.

Other types of pinch valves dispose with mechanical mechanisms entirely, the tube being squeezed shut by air or hydraulic pressure injected directly into the body of the valve (Figure 2). With a regulated fluid pressure the valve may

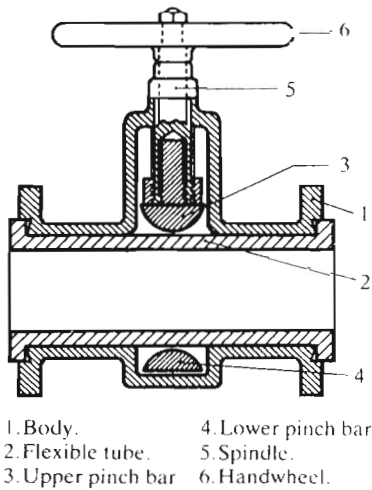


Figure 1. Pinch valve—basic components.

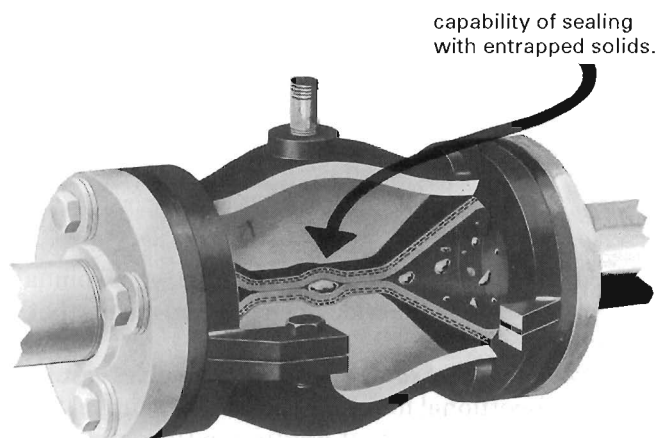


Figure 2. Hydraulic pressure-operated pinch valve.

be used for throttling as well as shut-off (full closure) (Figure 3). The particular advantage of the fluid-operated pinch valve is that it will still close tight over entrapped solids (making it particularly suitable for handling products with solids in suspension). Also, because the tube is flexed under uniformly distributed pressure, its life should be much longer than that of a similar tube working with a mechanical pinching system.

In common with the diaphragm valve, the operating mechanism is not in contact with the working fluid at any time, and nor is the body. In this respect, pinch valves have the advantage over diaphragm valves, unless the latter are rubber-lined or otherwise surface-protected. This exclusion of the working fluid from all parts excepting the sleeve itself makes it ideal for the handling of aggressive fluids and those which readily attack metal; and its straight-through characteristics mean it is suitable for the handling of slurries, pastes and semi-fluids generally, even those containing sizeable solid lumps in suspension.

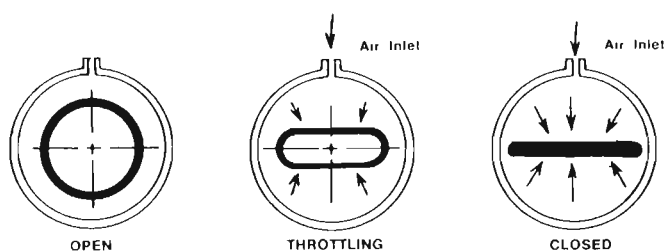


Figure 3.

Pinch valves with mechanical pinching mechanisms are normally operated by a handwheel and screw mechanism, but may equally well be driven by a powered actuator in larger sizes. Bodies are normally split horizontally to facilitate changing the tube when necessary without removing the complete valve from the pipeline.

Flow pattern

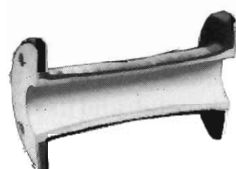
The flow pattern of a pinch valve is streamline and laminar. The non turbulent flow pattern even in the wide-open position means that wear on the valve sleeve is minimised.

The linear characteristics of some control pinch valves result in flow rates which are directly proportional to the amount of sleeve travel throughout the stroke of the valve while under constant-pressure and pressure-drop conditions.

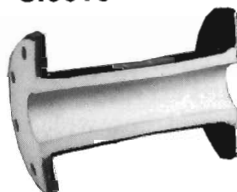
Typical valve sleeves are shown in Figure 4. These include standard, double wall for very abrasive conditions, cone sleeve for throttling control and variable orifice sleeve for improved flow characteristics where a high pressure drop is required.

Another form of pinch valve is shown in Figure 5. In addition to a resilient sleeve this also incorporates a streamlined core on to which the sleeve closes to seal. This valve can be operated in a variety of modes. In mode 1, the valve

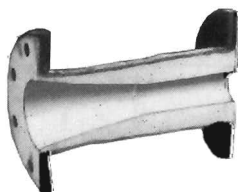
Standard Sleeve



Double Wall Sleeve



Cone Sleeve



Variable Orifice Sleeve

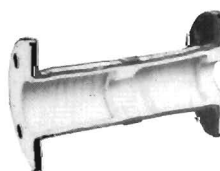


Figure 4. Typical control pinch valve sleeves.

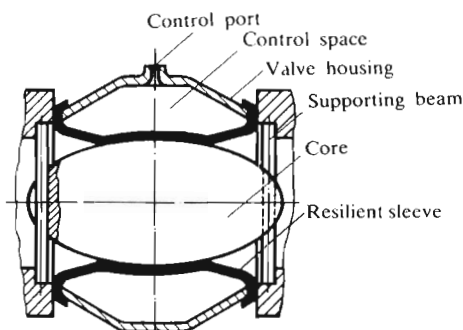


Figure 5.

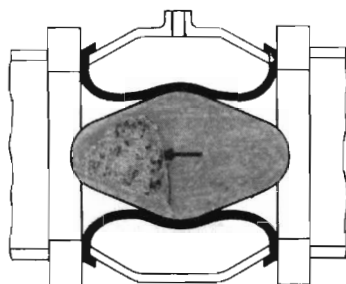


Figure 6.

remains closed when no pressure is present, due to the resilience of the sleeve (Figure 6). In the presence of pressure in the pipeline, the valve opens and remains open (Figure 7). It closes again in the absence of flow pressure. Line pressure of about 1 bar (14 lbf/in^2) is sufficient to hold the valve in the fully open position.

In mode 2, inlet pressure is tapped and fed to the control space between the sleeve and body, closing the valve. The valve opens when the operating pressure is relieved from the control space (Figure 8). At extremely low line pressures, circa 0.5 bar (7 lbf/in^2), the valve remains closed and drop-tight, even when operating pressure is relieved from the control space.

In mode 3, the operating principle is the same, except that the control space is pressurised from an independent source, e.g. compressed air or a hydraulic supply.

In mode 4, the flow is tapped to feed the control space which is only partially filled and then isolated, holding the sleeve in a partially closed position (Figure 9). In this mode the valve operates as a throttling valve or pressure-regulation valve.

Various low-hardness, high-tensile elastomeric compounds are used for the tubes, choice being made on chemical resistance and/or abrasion resistance required and service temperature. Typical materials used are:

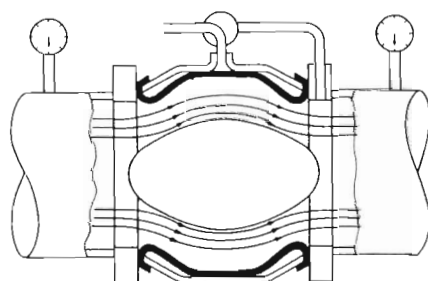


Figure 7.

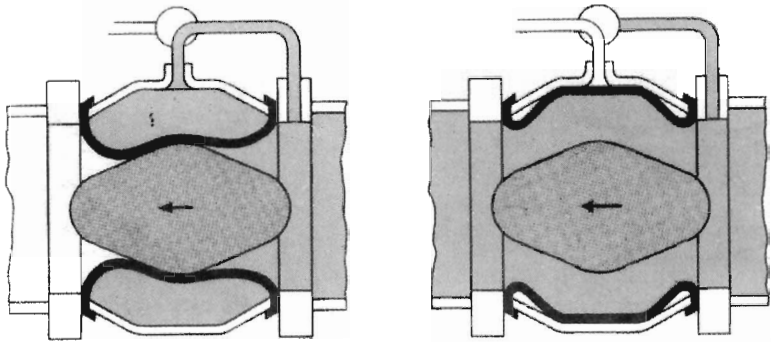


Figure 8.

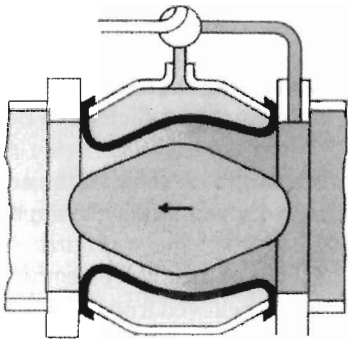


Figure 9.

GRS	excellent abrasion resistance.
BunaN	good resistance to solvents and hydrocarbons.
Neoprene*	good chemical and hydrocarbon resistance; in white compound also suitable for fast and dry application.
Butyl	good combination of chemical and temperature resistance; in white compound has good resistance to animal and vegetable oils.
Hypalon*	excellent chemical and temperature resistance.
PTFE	outstanding chemical and high-temperature resistance.
Silicone	good combination of high- and low-temperature resistance.
FDA	excellent abrasion resistance; suitable for many foods and dairy applications.
EPDM	excellent heat and chemical resistance.

*Neoprene and Hypalon are registered trademarks of DuPont Dow Elastomers.

Fluid performance

A pinch valve presents full-bore flow in the open position, with a straight, uninterrupted flow passage. Pressure drop or head loss under such conditions is thus minimal, related only to flow velocity and tube length, for a given fluid. There is no slamming when the valve closes against back pressure and the elastic nature of the tube tends to eliminate hammer (although this feature is absent in a hydraulically-operated pinch valve).

Vacuum services

Pinch valves tend to have limited suitability for vacuum services because of the tendency for the tube to collapse inwards. This is particularly true in the case of pinch valves with simple exposed tubes. Where the tube is enclosed in a body it is possible to adapt the valve for vacuum duties by applying a vacuum within the casing to balance the internal (vacuum) pressure.

Sizes and ratings

Sizes commonly available range up to 300 mm (12 in), with standard bodies suitable for pressures up to 14 bar (200 lbf/in²), or steel or stainless-steel

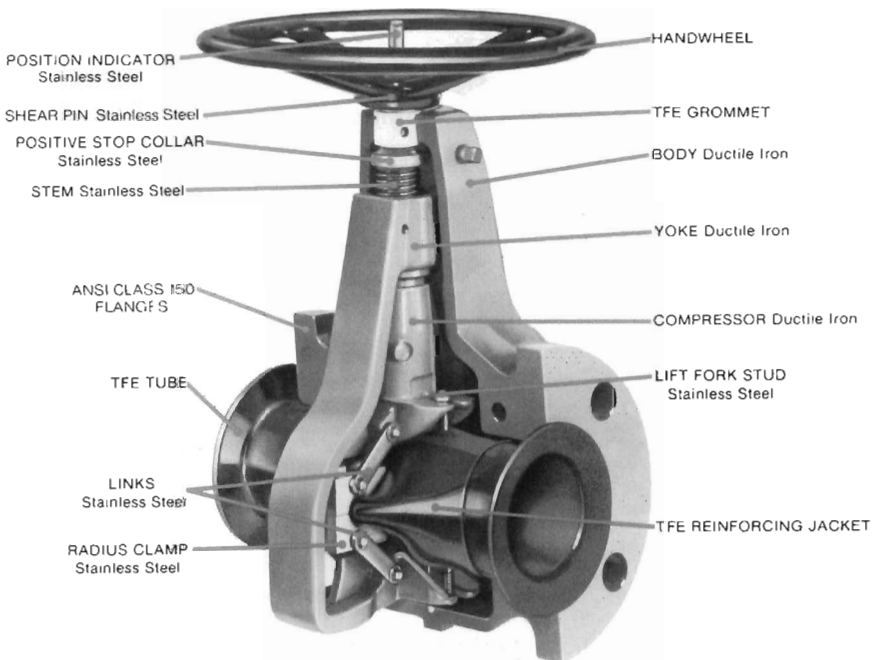


Figure 10. Clamp valve.

bodies for pressures up to 28 bar (400 lbf/in²). Larger sizes are also available, those over 600 mm (24 in) diameter usually being individually made with fabricated steel bodies.

Clamp valves

This type of pinch valve (Figure 10) consists of a flexible tube and clamp. The flexible tube has a heat-shrunk reinforcing jacket with both made from virgin TFE fluorocarbon resin. The clamping mechanism consists of a compressor which travels down a stem with rotation of a handwheel or power operator, and a yoke which travels up the stem at the same time.

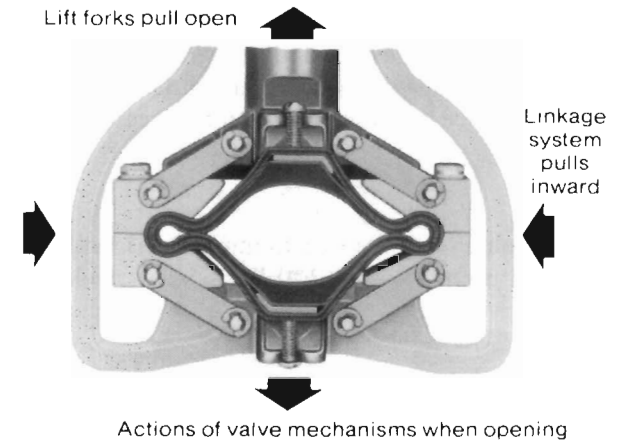


Figure 11.

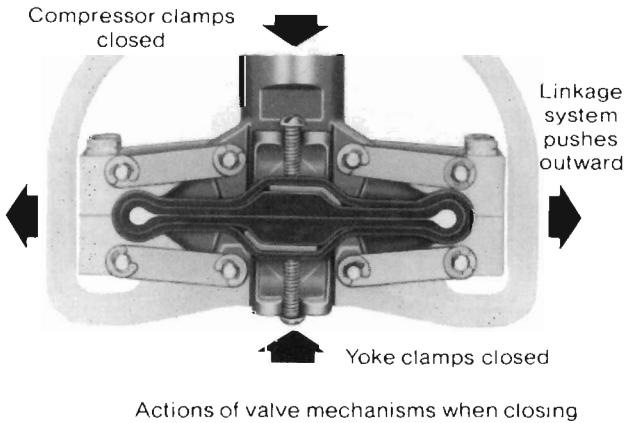


Figure 12.

Radius clamps are connected to the yoke and to the compressor by means of links and pins. Working together, they provide a 'scissor-jack' action which pushes the tube element inward during the opening cycle and pulls it outward during closing. The actions of the valve mechanism are shown in Figures 11 and 12.

Pinch valves are ideal for use in many industrial applications including:

- Chemical plants—where there are corrosive chemicals and for pump isolation.
- Power industry—FGD systems, ash handling and wet lime scrubbers.
- Mining industry—centrifuge control, solids separation, tailings systems, coal washing.
- Waste water treatment—flow equalisation, polymer feed systems, sludge control, grit systems, carbon slurry and raw sewage.

Diaphragm Valves

The distinguishing feature of a diaphragm valve is the closure device. This is usually an elastomeric diaphragm or tube which acts as a flexible seal when compressed against a ridge in the valve body. The diaphragm valve handles corrosive and erosive materials.

Diaphragm valves fall into two main types—*weir valves* and *straight-through valves*. In the former geometry (Figure 1), the body has a dividing weir, above which is mounted an elastomeric diaphragm. In the closed position the diaphragm sits on the weir. In the open position, it is fitted to provide a streamlined flow through the valve body. The amount of lifting is variable, so the valve can act both as a flow controller or a stop valve.

The straight-through diaphragm valve (Figure 2) may have a parallel, top-tapered or venturi-pattern body with closure provided by a wedge-shaped projection of the diaphragm. Because of the full-bore opening, it offers minimum resistance to flow in the open position, can pass suspended solids, and is capable of being rodded through for clearing any blockage.

Because the diaphragm isolates the moving parts, these valves are particularly suitable for handling aggressive fluids, as well as for 'clean fluid'

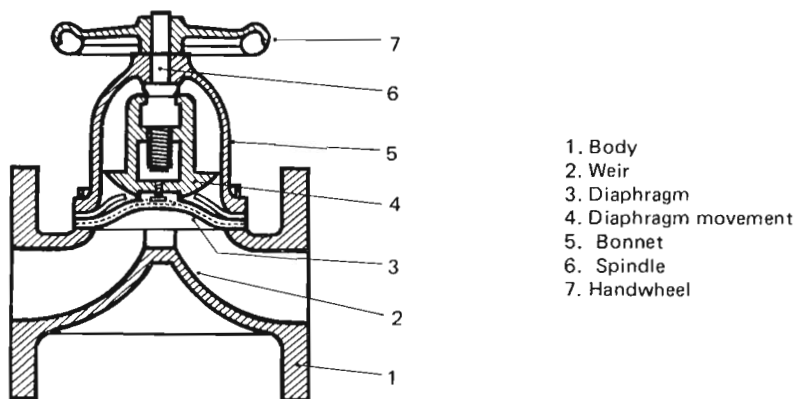


Figure 1. Weir-type diaphragm valve.

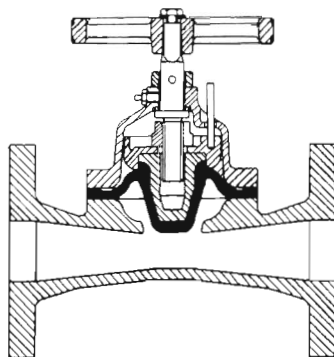
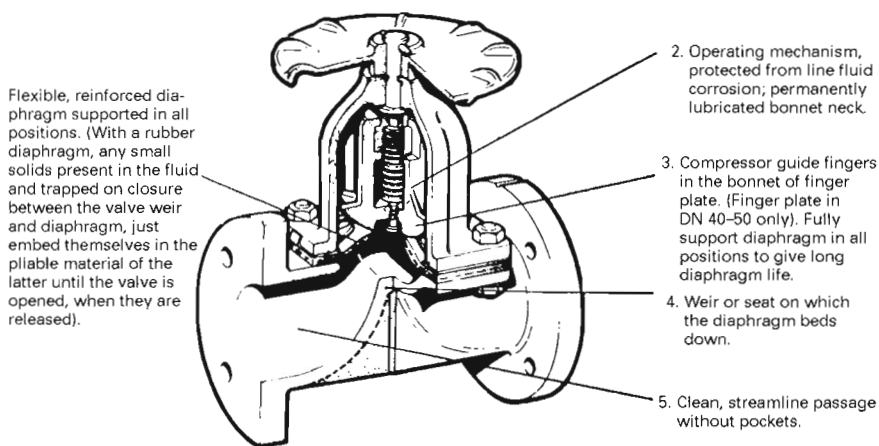


Figure 2. Straight-through diaphragm valve.



Standard diaphragm valve.

applications, the type of elastomer being chosen accordingly. The body itself can also be lined for corrosive duties.

Particular advantages of the diaphragm valve are the glandless construction and absence of seating problems. Its main limitation is that the maximum service temperature and pressure are limited by the temperature/pressure rating of the elastomeric material.

Typical body materials are cast iron, malleable iron, bronze, gun-metal and stainless steel. Lined diaphragm valves normally have cast-iron bodies lined with rubber, neoprene, polypropylene, PTFE or glass. Typical diaphragm materials and their main uses are summarised in Table 1. Reinforced diaphragm materials may be used for more arduous duties and are virtually standard for vacuum services.

Typical flow coefficients for weir-type and straight-through diaphragm valves are given in Tables 2A and 2B.

Table 1. Diaphragm materials

Material	Size		Temperature		Main uses
	mm	in	°C	°F	
Butyl rubber	15 to 350	0.6 to 14	−30 to 90	−22 to 134	Acids and alkalis
Nitrile rubber	15 to 350	0.6 to 14	−10 to 90	+14 to 134	Oils, fats and fuels
Neoprene	15 to 350	0.6 to 14	−20 to 90	−4 to 134	Oils, greases, air and radioactive fluids
Natural/synthetic rubber	15 to 350	0.6 to 14	−40 to 90	−40 to 134	Abrasives, brewing and dilute mineral acids
White natural rubber	15 to 125	0.6 to 5	−35 to 90	−31 to 134	Foods and pharmaceuticals White diaphragm
White butyl	15 to 150	0.6 to 6	−30 to 100	−22 to 212	Natural colour, foodstuffs plasticisers and pharmaceuticals
Viton*	15 to 350	0.6 to 14	+5 to 140	+41 to 284	Hydrocarbon acids, sulphuric and chlorine applications
Hypalon*	15 to 350	0.6 to 14	0 to 90	+32 to 134	Acid- and ozone-resistant
Butyl rubber	15 to 350	0.6 to 14	−20 to 120	−4 to 248	Hot water and intermittent steam services, sugar refining

*Du Pont Dow Elastomers Registered Trademark.

Operation

Of the standard forms of manual operation available, the handwheel is preferred for most purposes. The design of handwheel and operating mechanism (spindle, compressor, bush or nut) varies according to the size and type of valve. Various-shaped handwheels are also available. This facility, for instance, greatly assists the operator in identification in dark conditions or where the handwheel becomes slippery.

Various bonnets are applicable to handwheel-operated valves. Extended spindle valves are readily available but existing valves can be simply converted by use of adaptor and extension. Other variants of bonnet assembly can be substituted for the handwheel-operated design.

Normal bonnet assembly material is cast iron but alternative materials are offered by most manufacturers, e.g.:

- (i) All iron and steel construction (for handling acetylene, ammonia and other similar fluids).
- (ii) All stainless steel (where there is severe atmospheric-corrosion conditions).
- (iii) Gun-metal (for medical oxygen, turbine oils and certain water services).
- (iv) Cast steel with rising spindle fitting (mainly used in oil refineries).

Table 2A. Typical flow coefficients for weir-type valves

Valve size	Cast iron		Rubber-lined		Halav/glass-lined	
DN	C _v	K _v	C _v	K _v	C _v	K _v
15	5.8	1.6	1.3	4.4	6	1.7
20	11.5	3.2	9.4	2.6	12	3.4
25	17.4	5	14.5	4	18	5.6
32	26.5	7.5	22	6	28	8
40	43	12.2	35	9.8	45.5	12.8
50	84	21.4	70	17	88	22.5
65	126	30.5	102	24.4	132	32
80	180	44.4	147	35.5	185	46.6
100	320	77.7	264	62	336	81.6
125	420	108	348	86.4	444	113
150	600	147	504	117.6	630	154
200	1260	305	996	244	1320	320
225	1630	388	1320	310	1680	407
250	1990	500	1620	400	2076	525
300	2580	625	2088	500	2700	656
350	3840	833	3060	666	4020	875.7

Table 2B. Flow coefficients for typical straight-through valves

Valve size	Cast iron		Rubber-lined		Halav/glass-lined	
DN	C _v	K _v	C _v	K _v	C _v	K _v
15	8.6	1.8	—	—	9	1.9
20	—	—	—	—	—	—
25	37.8	7.9	30.6	6.5	39	8.2
32	55.8	12.5	45.6	10.1	58.2	12.9
40	75	20	66	18	79	21
50	128	34	107	28	138	36
65	238	68	195	55	254	72
80	330	90	264	73	342	92
100	588	150	480	123	618	158
125	924	235	720	184	960	274
150	1680	385	1260	293	1800	545
200	2580	789	2196	665	2724	825
250	4020	1050	3420	900	4296	1100
300	6060	1510	4884	1221	6204	1550
350	10.300	2800	9950	2360	—	—

- (v) Silicon aluminium (for lightness and low temperature).
- (vi) A variety of plastic materials (for general corrosion resistance).
- (vii) Epoxy-coated cast iron (for corrosion resistance, attractive appearance).

For quick closure and/or opening, lever-action valves can be used. A quarter-turn movement is preferred when the liner can also act as a valve

position indicator, e.g. parallel with the pipeline in the fully open position and at right angles to the pipeline in the fully closed position.

Diaphragm valves are particularly suited for rapid closure/opening because the cushioning action of the flexible diaphragm minimizes the shock throughout the pipeline, compared with equally rapid but less resilient types of valve movement.

Double-diaphragm valves

A double-diaphragm valve is a basically simple design consisting of two flexing diaphragms on an axial stem. There is no metal-to-metal contact, O-rings or sliding seals, making lubrication unnecessary, as shown in Figure 3. The valve requires no special filtration and is tolerant to dirt-laden media as well as scrubbed instrument air.

Designed for compressed air and inert gases, the self-purging non-lubricated design is also suitable for vacuum applications. A cartridge-insert version is manufactured for direct integration into finished products. Typical areas of application include food processing and medical equipment. In fact, a double-diaphragm air valve has performed over two billion cumulative cycles, an equivalent of 50 years of human life, in various heart systems. The valves are located in the mechanical control system, which is attached to the patient by plastic tubes, and control the flow of compressed air through the tubes to the artificial heart, causing it to pump.

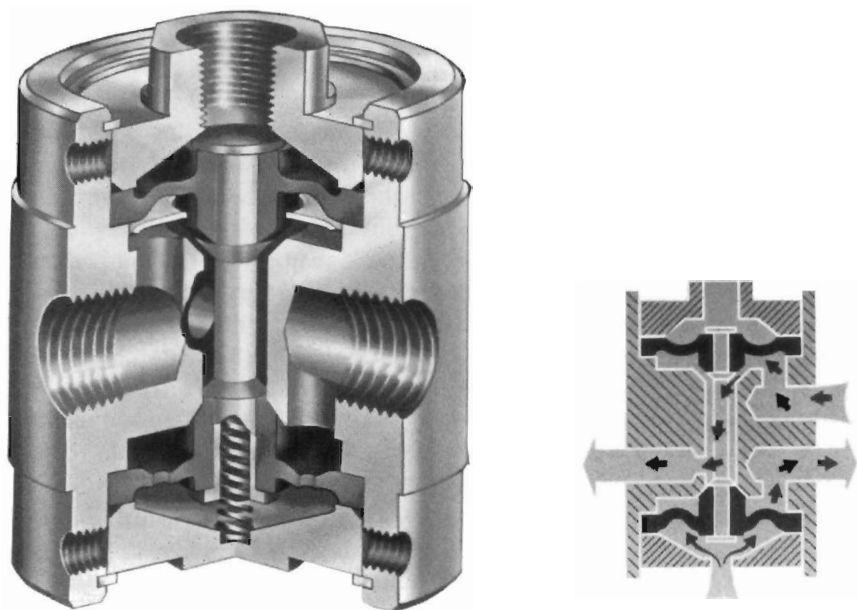


Figure 3. Double-diaphragm air valve.

Plastic diaphragm valves (Figure 3) are becoming increasingly popular. Manufactured from PVC-U, PVC-C, ABS, PP and PVDF plastics, the main advantages are lightweight and compact construction, corrosion resistance and a long maintenance-free working life.

Typical pressure–temperature charts are shown in Figures 4 and 5. Miniature air-operated diaphragm valves are designed for pressure applications for

PVC-U, PVC-C, ABS

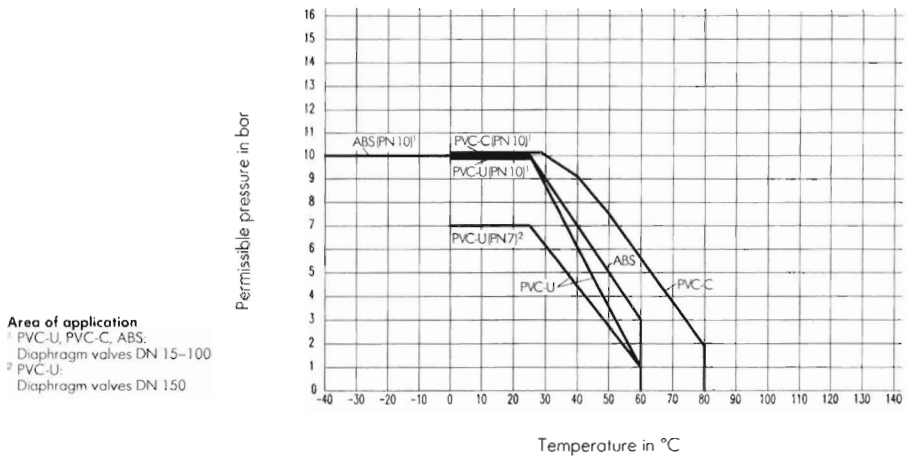


Figure 4. and 5. Typical pressure–temperature diagrams for diaphragm valves.

PVC-U, PVC-C, ABS

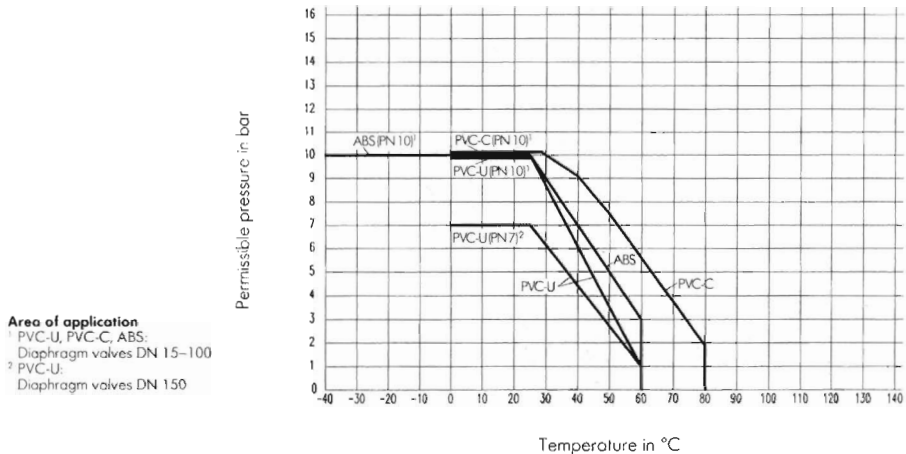


Figure 4. and 5. Typical pressure–temperature diagrams for diaphragm valves.



Figure 6. Iris-type diaphragm valve.

highly corrosive or ultra-pure liquids, when space is critical. This type of valve has numerous applications in the electronics/semiconductor industry for use with concentrated etchants and ultra-pure water.

Generally, the valves open with air pressure and close with a return spring.

Diaphragm valves play a significant role in air-pulse jet-dust control equipment and valve performance has a great influence on the cleaning efficiency of the generated air pulse.

Air-pulse diaphragm valves need to open and close very rapidly at high flows. Fluttering of the diaphragm during opening and closing increases air consumption and will affect valve performance. The opening time of the pulse valve must be as short as possible, i.e. between 8 and 14 ms to achieve best performance. Long closing times increase air consumption.

Iris-type diaphragm valves

The diaphragm valve shown in Figure 6 is designed specifically to control the flow of dry bulk solids and is used for food, chemical pharmaceutical and general processes. Essentially the valve consists of a tube of flexible material, fixed rigidly at each end. This tube is held in a mechanism which rotates one end through 180° relative to the other. When twisted in this way, the tube closes completely, creating a diaphragm visually resembling the iris of a camera.

Iris-type diaphragm valves have a variable area concentric orifice so that the flow can be accurately controlled. They can open or close irrespective of size to the smallest of orifice sizes, regulating the flow of even the finest of powders. Power-operated models can open or close to pre-set "dribble-feed" positions.

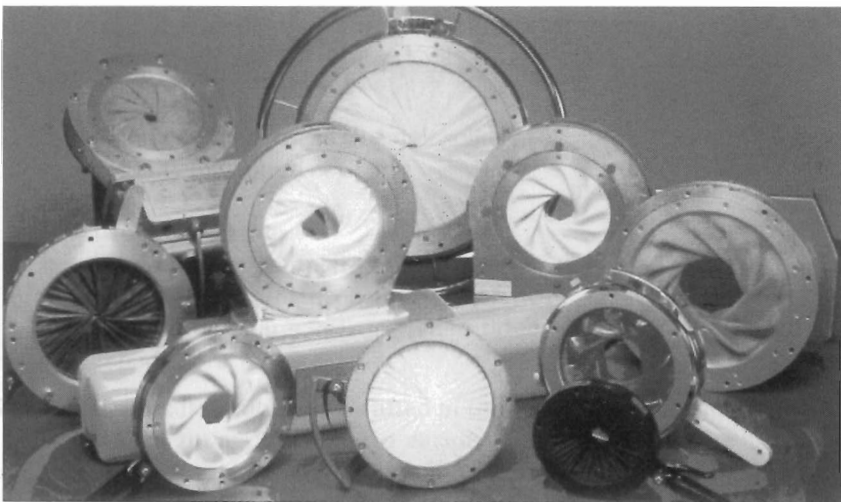


Figure 7. Iris-type diaphragm valve membranes.

Central to the performance of the iris-type diaphragm valve is the correct selection of diaphragms.

The diaphragms are usually made from either (i) elastomers—natural and synthetic, (ii) fabrics and (iii) coated fabrics (Figure 7).

Typical applications include discharge valve to cake-mix mixer, discharge valve for rail tankers carrying bulk sugar, packing herbs and spices, controlling the flow of finished tablets and capsules, general purpose flow-control valves for silos and hoppers, filling fertilizers into 1 tonne bags, gland around telephone cable to remove excess grease, valve on salt silo for motorway gritting, weighing out of abrasive ingredients, and controlling the flow of pigments.

Slide Valves

Slide valves are one of the many variations of the gate-type valve and were originally designed for use in the pulp and paper industry to overcome the problems in handling wet and dry fluidized solids in pipelines.

The slide valve, or knife or plate valve as it is sometimes called, in its original form was a simple rattle-fit plate in a fabricated body. Opening or closing was achieved by pushing or pulling the plate to give a crude shut-off. Although design improvements have been made, this simple form of valve is still commonly used as a diverter or hopper outlet valve on dry powders or solids where a pressure seal is not required. In such applications the body of the valve will probably be of square or rectangular bore to suit the hopper outlet or duct on which it is fitted.

The logical advancement from this design was the fitting of resilient seals and an operating screw and handwheel to open and close the slide. With the improvement in sealing that this gives, the slide valve is suitable for a much wider range of services and can handle solids suspended in liquids or gases. Although they are simple valves, there are a variety of designs on the market all sharing, to a greater or lesser degree, the following advantages:

- (a) short overall length
- (b) no wedging action
- (c) thin closing member to cut through solids in the line
- (d) substantially full-round bore
- (e) lightweight
- (f) easy to power-operate
- (g) stem/slide connection not in contact with line fluid
- (h) slide exposed for visual inspection when valve is open
- (i) good regulating characteristics when fitted with specially-shaped slide and/or body bore

The resilient seals, which can be fitted in the body or on the slide depending on individual design, are available in various materials to suit the fluid being handled. For most applications they will be in rubber, e.g. Nitrile or Viton*.

*Dupont Dow Elastomers registered Trade Mark.

but for higher temperatures or in the food and chemical industries it may be necessary to specify PTFE or other special elastomers.

There are many individual designs of slide valve on the market. In some ways, this is advantageous as it gives the valve user more scope for finding a valve specifically designed to overcome a unique problem. Generally speaking, slide valves are used on services such as:

- (i) viscous media
- (ii) dry powder
- (iii) dry solids of small and large particle size
- (iv) slurries and sludges

These applications are commonly found in the waterworks, sewage, mining, chemical, power generation, food, cement, brewing and other process industries as well as the pulp and paper industry for which the valve was originally designed. But this does not necessarily mean that slide valves are only used on difficult applications. They are equally successful when put on less arduous duties where the benefits of leak-tightness, space-saving, price, etc., need to be considered.

Figure 1 shows a typical design of slide valve. It will be noted that the bore is clear of obstructions and that there are no pockets in which solids can collect,

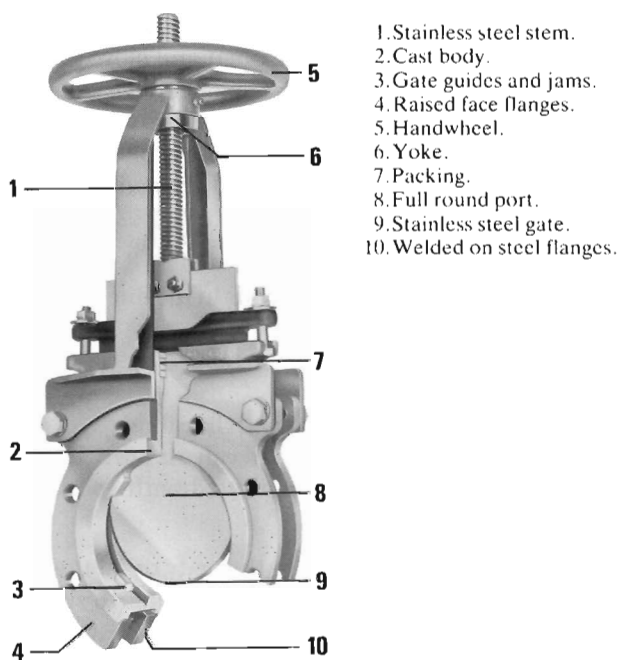


Figure 1. Knife-edge slide valve.

in addition to which the seal is housed in the body of the valve and shrouded by the body to avoid damage from solids in the line. In this design the seal seats on the edges of the slide, which means that the valve will seal equally well with flow in either direction and the slide has a chamfered leading edge which enables it to slice through any obstruction and seat effectively against the body seal (Figure 2). These features ensure bubble-tight shut-off against pressure or vacuum.

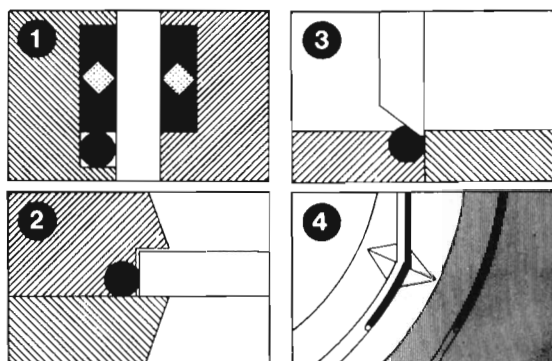


Figure 2. Features of the slide valve shown in Figure 1. 1. Built-in horizontal seal; 2. internal seal located in valve body; 3. valve slide with chamfered end; 4. internal contours designed to ensure that deposits are flushed into the flow.

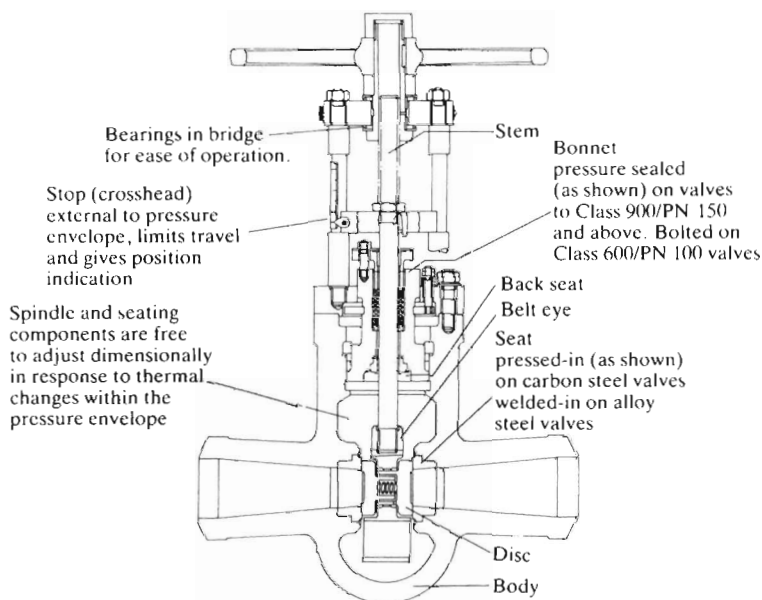


Figure 3. Venturi parallel-slide valve.

Unlike conventional gate valves, the slide passes through the body and a standard gland packing cannot therefore be used. In the design shown here, profiled transverse seals are employed. These comprise two lengths of V-section rubber with the open sides of the V facing each other. During assembly, the diamond-shaped opening is filled with an oiled-fibre packing which is forced

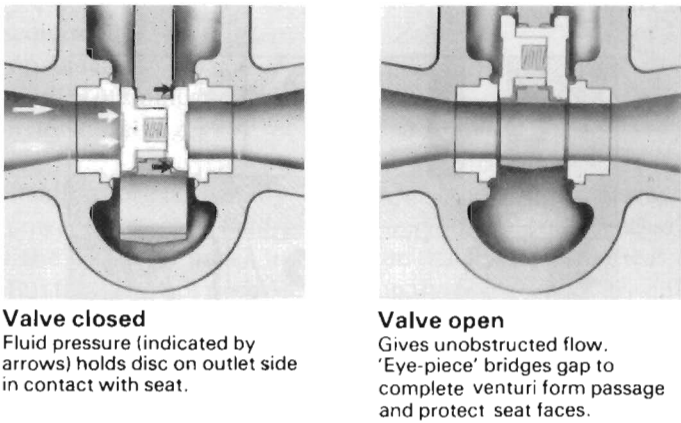


Figure 4. Parallel-slide action.

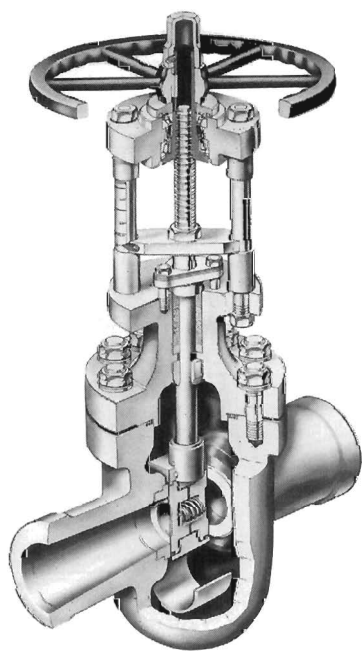


Figure 5. Parallel-slide valve.

in and pushes the lips of the transverse seals into contact with the body and the slide. In this way, leakage to atmosphere is prevented. Also, by removing the screws on the sides of the body, more packing can be inserted at a later date to maintain a tight seal even if the valve is on-stream and under pressure or vacuum. An optional extra on this design of valve is a set of scraper blades set below the transverse seals. During the opening phase, this enables the slide faces to be scraped clean over their whole width before they are drawn through the transverse seals, and damage to the seals is thereby prevented. This is particularly important when handling sugar, sticky media such as honey, chocolate compound, glues, tenacious powders, etc.

A venturi parallel-slide valve is shown in sectioned form in Figure 3. This valve incorporates a pressure-sealed bonnet design and is used as a general purpose stop valve for main steam and feedwater isolation on power station boiler plant. The parallel-slide action is shown in Figure 4.

A cutaway view of the internal cross-section of the valve is shown in Figure 5.

When power operation is required, slide valves are particularly easy to operate pneumatically as the pneumatic cylinder can be mounted directly on



Figure 6. Slide valves used in (top left) brewing, (top right) food processing and (bottom) dust control.

the valve pillars and the end of the piston rod connected direct to the valve slide. Electric actuation is also widely used on slide valves and, to a lesser extent, hydraulic actuation. In addition, a very useful alternative to handwheel operation is the lever-operated slide valve which gives a very quick open/close operation.

When powders and granular materials are stored in silos or conveyed by belt and screw conveyors, actuated rectangular valves are often required, as shown in Figure 6. This is particularly true for pulverised fuels, ash and grain, as well as brewing, food processing and dust control.

See also the chapter on Gate Valves.

Screw-down Valves

The general classification "screw-down valve" is taken to refer to all types of valves sealing by a disc or plug, etc., and in which the sealing element is lifted from and lowered on to the valve seat by rotation of a threaded stem, the axis of which is perpendicular to the valve seat. Mainly this embraces various types of stop valves, e.g. globe valves, oblique or Y-valves, gate valves, lift-type plug valves and angle valves. Much of this has been covered in other chapters. It also includes certain types of throttling valves, i.e. needle valves in particular.

Screw-down valves are also categorised as:

- (i) Inside screw, where the threaded portion of the stem is fully enclosed within the bonnet.
- (ii) Outside screw, where the threaded portion of the stem is exterior to the bonnet and (usually) carried in a yoke (see Figure 1).

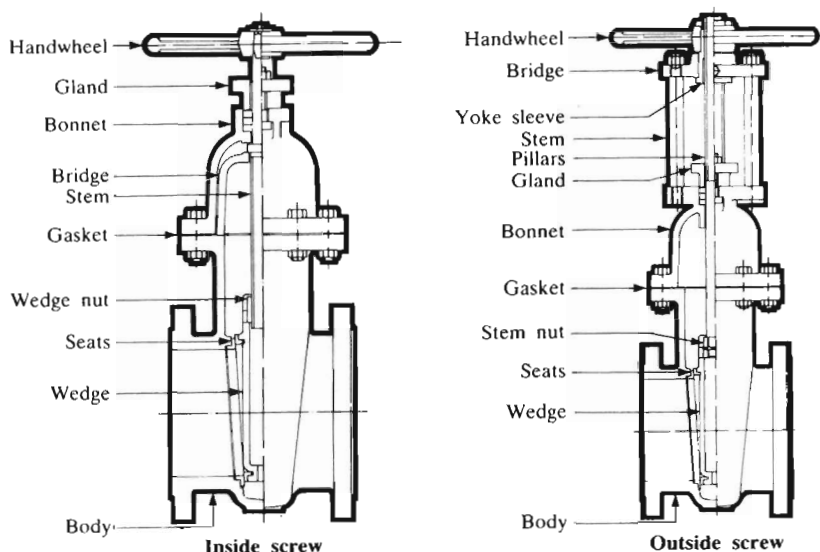


Figure 1. Outside- and inside-screw gate valves.

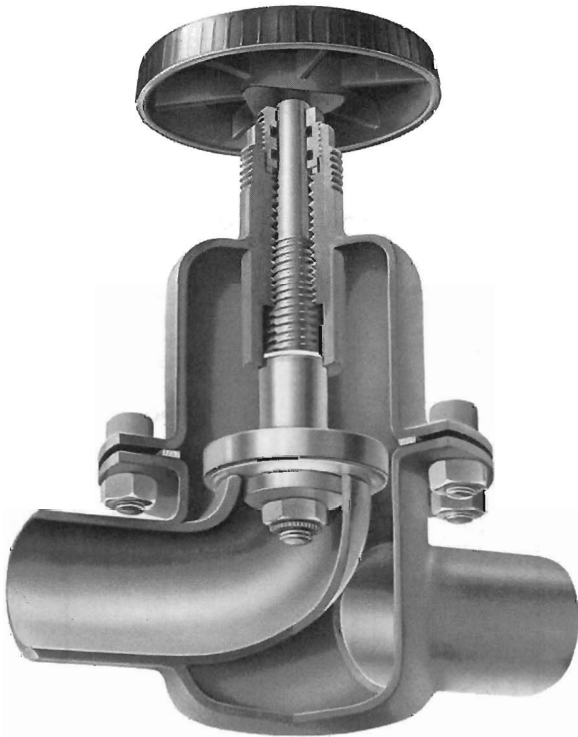
A further distinction is made between:

- (a) Rising-stem valves, where the stem moves in or out of the bonnet as the stem is rotated by a handwheel, lever or actuator.
- (b) Non-rising stem valves, where there is no displacement of the stem along its axis when rotated.

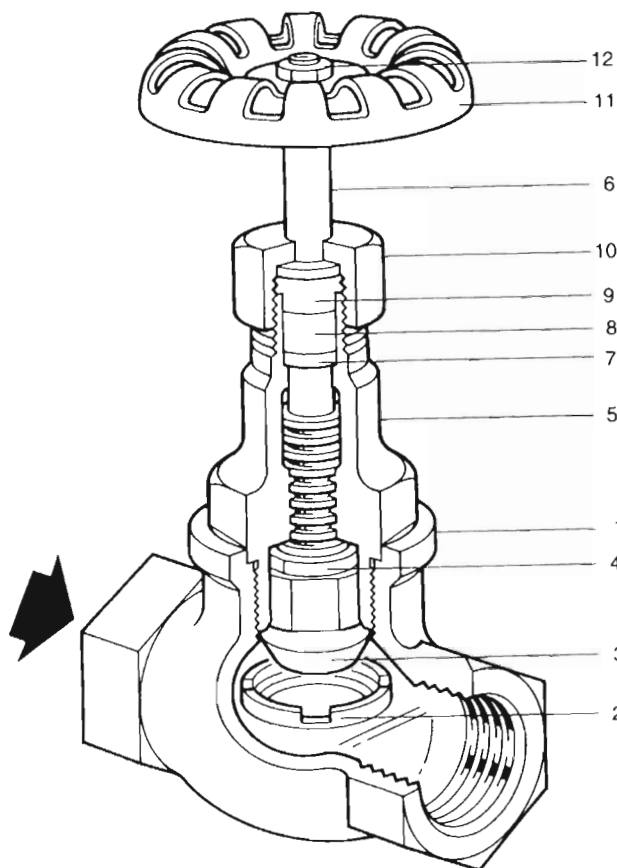
Both inside-screw and outside-screw valves can be of rising-stem or non-rising-stem type.

Principal differences between these categories are:

- (i) Inside-screw valves have the threaded length of the stem protected from dirt, etc. However, the stem is fully exposed to the fluid being handled. Also it is more difficult to provide lubrication for the thread where the fluid handled is not itself a lubricant.
- (ii) Outside-screw valves have the threaded length fully exposed to the surroundings, hence they can readily collect dirt and/or be subject to corrosion. Lubrication is easier because the threaded length is fully

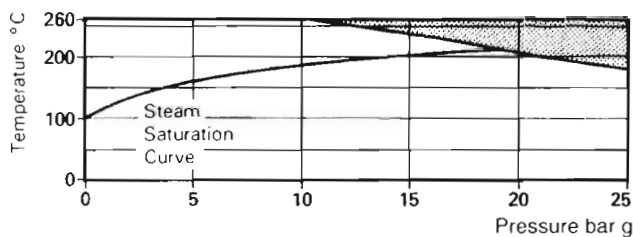


Inside screw-down valve.



No	Part
1	Body
2	Valve Seat
3	Valve
4	Locknut
5	Bonnet
6	Stem $\frac{1}{2}$ " - 1" $1\frac{1}{4}$ " - 2"
7	Washer
8	Gland Packing
9	Gland
10	Packing Nut
11	Handwheel
12	Handwheel Nut

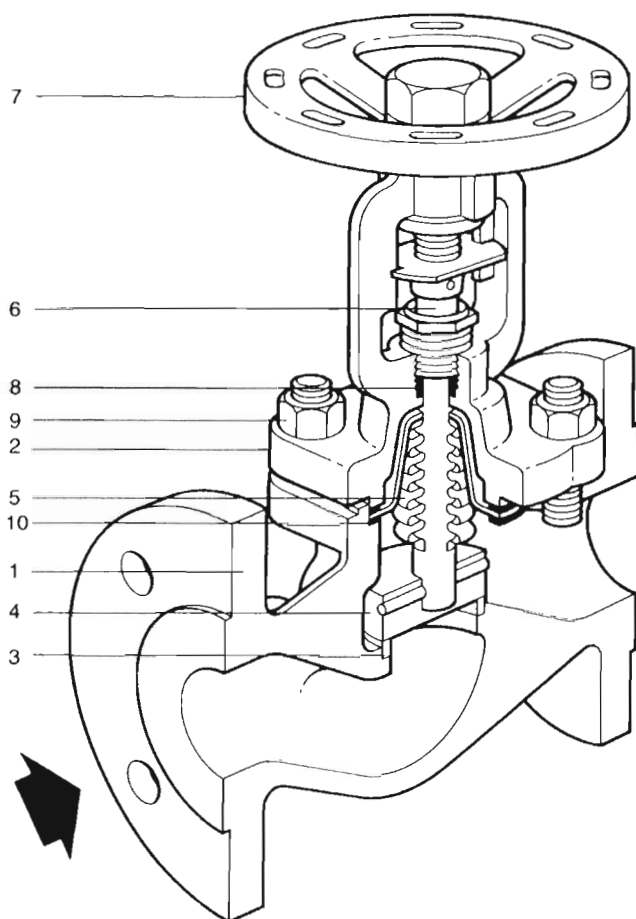
Operating range



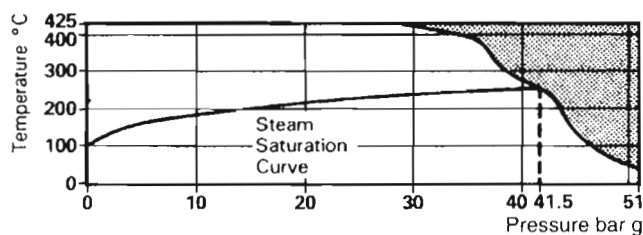
 The product must not be used in this region.

Typical industry standard stop valve.

No	Part
1	Body
2	Bonnet
3	Seat
4	Disc/Plug
5	Bellows
6	Stem
7	Handwheel
8	Stem Packing
9	Bonnet Studs Bonnet Nuts
10	Body/ Bonnet Gasket

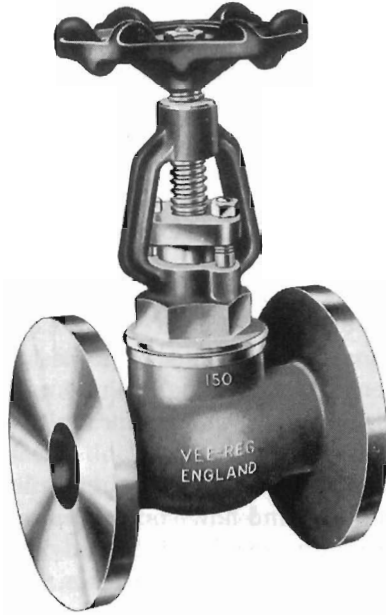


Operating range (ANSI 300)



 This product must not be used in this region.

ANSI bellows-sealed stop valve.



Outside-screw rising-stem stop valve.

accessible, but again lubricant will promote collection of dirt on the threads. The threaded length of the stem is isolated from the fluid being handled by the stem glands, hence this type is more suitable for handling corrosive media, slurries, etc.

- (a) Rising-stem valves provide a visual indication of the position of the valve disc or gate and thus the degree of valve opening. On the other hand, adequate clearance is needed above the valve to accommodate the rising-stem movement.
- (b) Non-rising-stem valves have the advantage that they can be installed in positions where headroom is limited. Stem wear is also minimised, although there may be increased wear on the threaded length which lifts and lowers the valve disc or gate.

For corrosive duties, stem protection may be provided by a bellows seal applied either to an inside screw (usually) or an outside screw.

Spool Valves

Spool valves embody two basic elements: a cylindrical barrel in which slides a plunger or spool. Port blocking is provided by glands or full diameter sections on the spool, with intervening waist sections which provide port interconnection through the barrel. This makes it easy to provide multi-way and multi-position switching.

Sliding-spool valves are generally used in hydraulic and pneumatic fluid power systems for directional and flow-control purposes. There must be an annular clearance between the spool and the body, so there is always some leakage flow across the spool and this must be taken into consideration.

Spool valves are relatively simple and economic to manufacture, although for adequate sealing a fine surface finish is required on both the spool and the barrel bore, with close tolerances to ensure practical minimum clearances. Glandless spool valves thus normally require a lapped fit between spool and body.

Pneumatic-spool valves with static seals offer simpler construction in this respect and also rather more flexibility in design with seals positioned between valve spaces so that a seal is situated between each subsequent port and one seal on the outside of each of the two outer ports (see Figure 1).

Spool valves operate on a sliding principle, so design normally follows the basic requirement of all slide valves, i.e.:

- (i) Pressure-balanced ports are required so that there is no net pressure force acting axially on the spool.
- (ii) The valve diameter should be a minimum consistent with suitable stiffness.

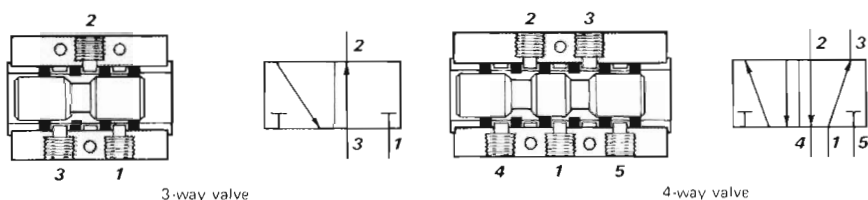


Figure 1. 3-way and 4-way spool valves.

- (iii) The valve body or sleeve must have adequate rigidity.
- (iv) Friction forces must be minimised and are largely controlled by material selection for rubbing/sliding parts.
- (v) Annular flow should be symmetrical in order to avoid radial unbalanced forces which could increase friction.
- (vi) Bernoulli forces, arising from changes in fluid momentum, must be minimised.

Parameters (v) and (vi) are largely controlled by the detail design of the spool.

Spool-cushioning passages can be built into the valve as shown in Figure 2. These equalise hydraulic forces on the ends of the spool and cushion the spool shift. When the spool is shifted, the fluid displaced from one end of the spool is transferred to the other end through the passage which is designed to provide a cushioning effect and balance the spool.

Forces may also be set up due to the changes in fluid momentum through the valve, generally described as Bernoulli forces. Thus, typically, there may be a reduction in pressure on the valve spool at the controlling edge, leading to a force being generated producing unbalance or tending to close the valve. At the same time, if backlash is also present in the system, Bernoulli forces may produce high frequency 'chatter' of the valve spool.

The hydraulic unbalancing effects of fluid momentum between the cylinder and tank ports of a valve can be minimised by contouring the spool shape as shown in Figure 3. Flow forces that are developed at the conventional square land orifice (P to B) are partially compensated for by the force-balancing contour on the outer spool lands (A to T).

Accurate sequencing of land opening and closing also provides maximum axial stability, as shown in Figure 4. In this example, it is important that flow path A to T is opened before the path P to B to prevent pressure intensification which could upset axial balance and limit valve function.

The spool can be moved manually, mechanically, by pilot pressure or by an electric solenoid. Directional-control valves usually have finite spool positioning to change the direction of flow from one port to another.

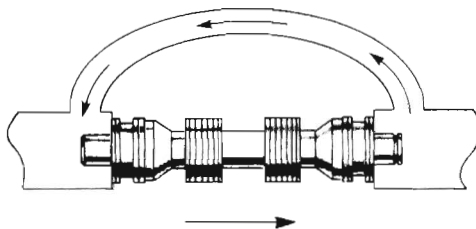


Figure 2.

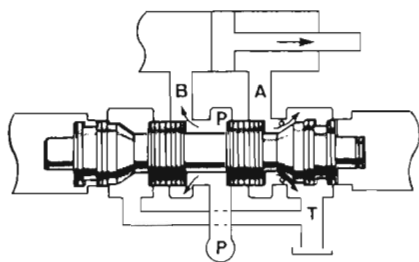


Figure 3.

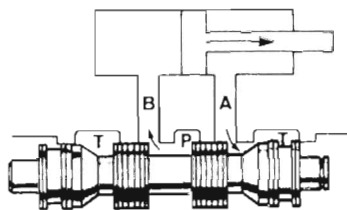


Figure 4.

Proportional-control valves and servo-valves have infinitely controllable spool positioning so that the flow rate of the fluid as well as the direction of flow is controlled.

Servo-valves

A servo-valve is capable of providing continuously variable flow with changing input signal. The latter is normally an electric signal to a torque motor, but feedback signals may also be derived hydraulically or mechanically.

The simplest form of spool valve to control both the direction of flow and flow rate is the three-way valve, but this will control the flow to only one side of the load and a differential-area spool must be used. A more common design, shown in Figure 5, is the four-way valve whose symmetry results in improved linearity. However, there are three critical axial dimensions of the spool and the sleeve. To reduce some of the manufacturing difficulties, designers have used the following different designs.

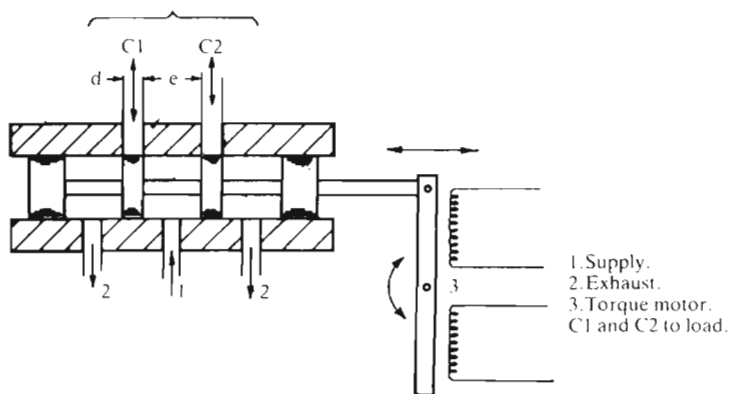


Figure 5. Spool-type valve.

Split-spool valves

To reduce the number of critical axial dimensions of the four-way valve, two separate three-way valves may be linked together, as shown in Figure 6, and a means of zero adjustment can be provided in the links. Although manufacturing problems associated with porting are reduced, further difficulties are introduced in that two parallel bores have to be made, and there may be additional backlash in the linkage. Furthermore, its weight is usually more than that of the simple four-way valve.

A design in which manufacturing difficulties associated with axial tolerances are eliminated is the Elliott adjustable lap valve. This is a split, three-way valve in which the two spools are mounted back-to-back and are actuated by a central ram against restraint. The correct valve lap is obtained after assembly by axial adjustment of the sleeves.

Sliding-plate valves

This design, shown in Figure 7, may be likened to an unwrapped spool (that is, two-dimensional), or even to the original D-type steam valve. It overcomes the difficulties associated with the manufacture of the bores in spool valves, and hole-and-plug porting techniques may be used. However, some manufacturers consider, the difficulties associated with the production of flat and parallel plates greater than those of making spools and sleeves. Various methods are used to reduce friction forces; in some valves the sliding member is suspended on spring plates to prevent metallic contact, and others achieve hydrostatic pressure balance.

Rotary-plate valves

An alternative form of the plate valve is the rotary type shown in Figure 8. Reaction-force compensation may be fairly easily introduced by the use of deflector vanes which have the added advantage of being adjustable.

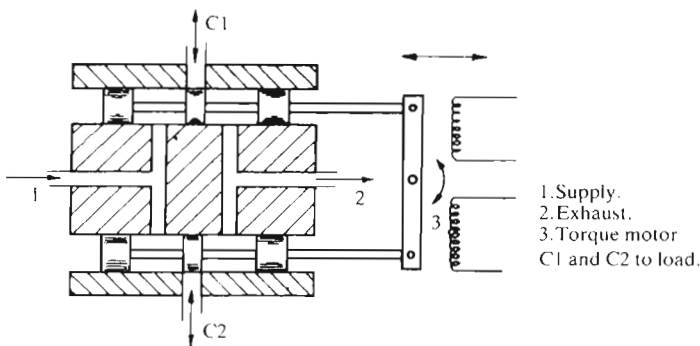
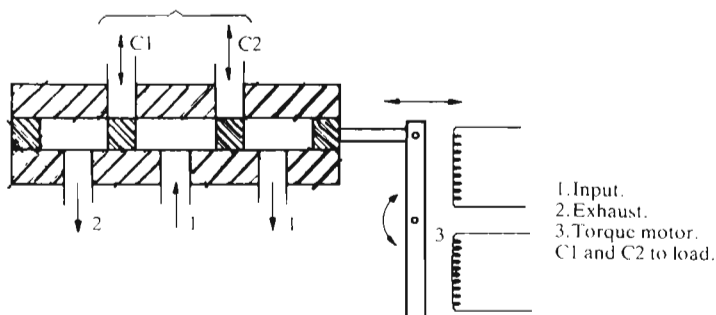
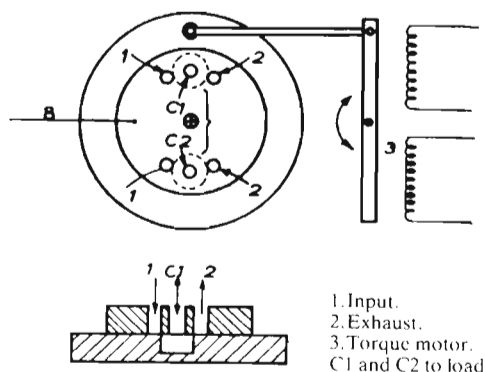


Figure 6. Split-spool valve.

Figure 7. *Sliding-plate valve.*Figure 8. *Rotary-plate valve.*

Askania-type valves

The main advantages claimed for this type of valve, shown in Figure 9, are that it is less susceptible to contamination clogging, and the ease of manufacture. Although it has been fairly extensively used for control purposes in low-pressure applications, its design for medium- and high-pressure systems presents a far more difficult problem because it is based on empirical methods. A comment has been made that large reaction forces leading to serious instability can occur, although the reasons for this are not known. This valve, whose action depends upon the conversion of kinetic energy of the jet into static pressure at the spool or ram, is referred to in the United States as the 'jet-pipe' design. However, it is probably preferable to avoid this term to prevent confusion with other nozzle or jet designs used in two-stage valves.

Two-stage valves

Two-stage valves were designed to overcome the practical limitations of power and response of single-stage valves. Any of the previously mentioned

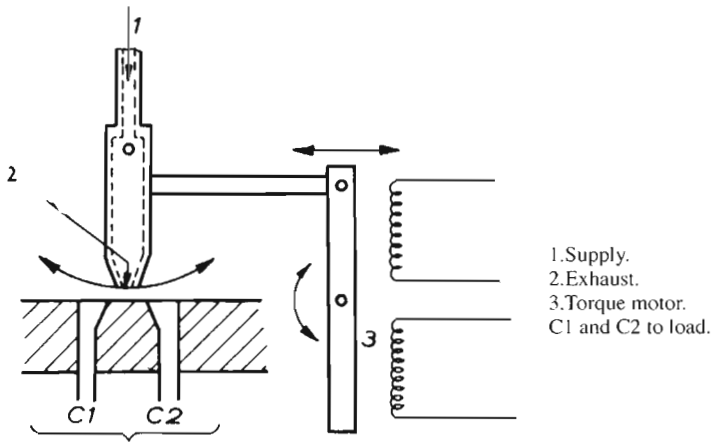


Figure 9. Askania-type spool valve.

single-stage valves may be used as the first stage of a two-stage valve with a spool or a plate as the second stage, of which the following are examples.

Double-spool valves

In this valve, shown in Figure 10, the output from the first stage is used to control the motion of the second stage spool and hence produces delivery to the load. Feedback from the output stage is necessary to avoid the integrating effect of the two spools and is usually accomplished either electrically or mechanically.

Nozzle-flapper spool valves

The basic configuration of the single nozzle-flapper valve is shown in Figure 11, where it can be seen that a movement of the flapper towards the nozzle

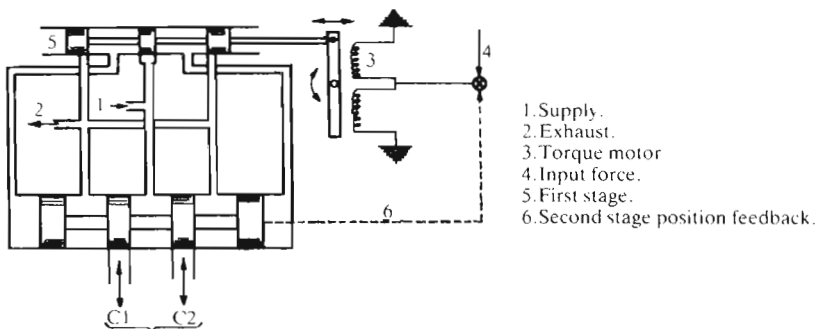


Figure 10. Two-stage valve: double spool.

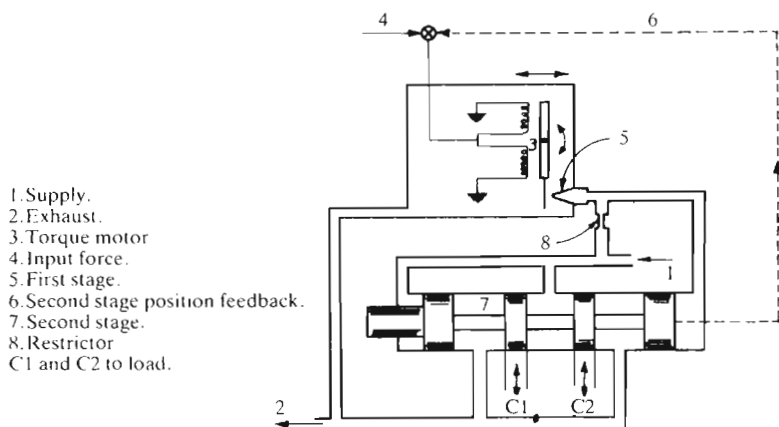


Figure 11. Two-stage valve: nozzle-flapper spool.

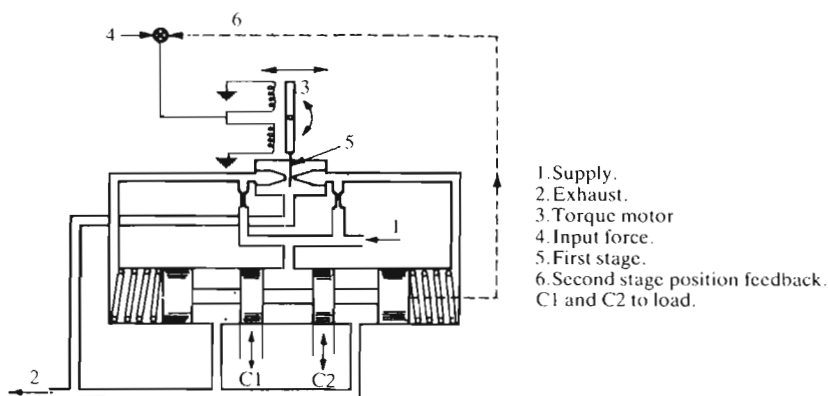


Figure 12. Two-stage valve: double nozzle-flapper spool.

causes an increase in the pressure in the chamber between the nozzle and the restrictor which causes the second stage spool to move and thus deliver oil to the load. Movement of the flapper in the other direction will reduce the chamber pressure, causing the spool to move in the opposite direction and produce reverse motion of the load. Thus the nozzle and flapper act as a variable impedance, and a variable percentage of the supply pressure acts on the right-hand end of the second stage spool.

Double nozzle-flapper spool valves

Representing a further development of the design previously described, this valve is shown in Figure 12. The flapper and nozzle control the pressure at

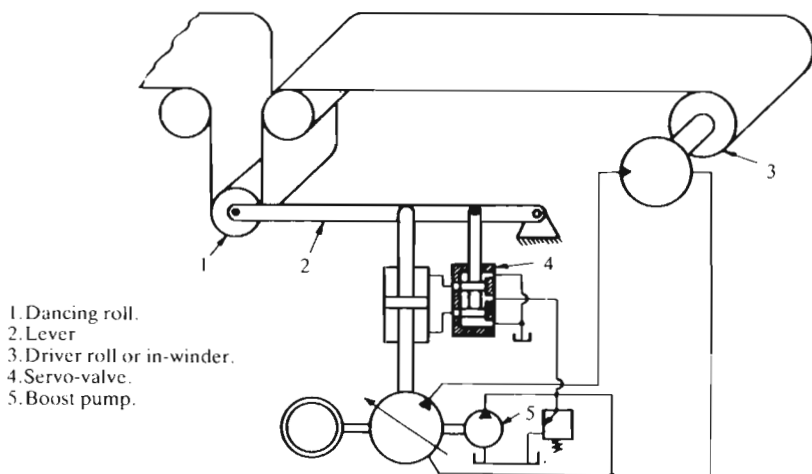


Figure 13. Controlling web tension with hydraulic servo-operated variable delivery pump and hydraulic motor drive.

both ends of the second stage spool and thus the operation of the first stage may be likened to that of the four-way valve. Although a fairly high quiescent flow is inevitable with this design, the power loss is not significant for most applications.

The introduction of the nozzle-flapper device as the first stage, with its low inertia and short stroke, was a major contribution in the field of two-stage valve design. The design shown in Figure 13 is basically one of the earliest of a family of valves having nozzle-flapper as the first stage.

For more detailed information on spool valves, refer to the *Pneumatic Handbook* also published by Elsevier Science Limited.

Solenoid Valves

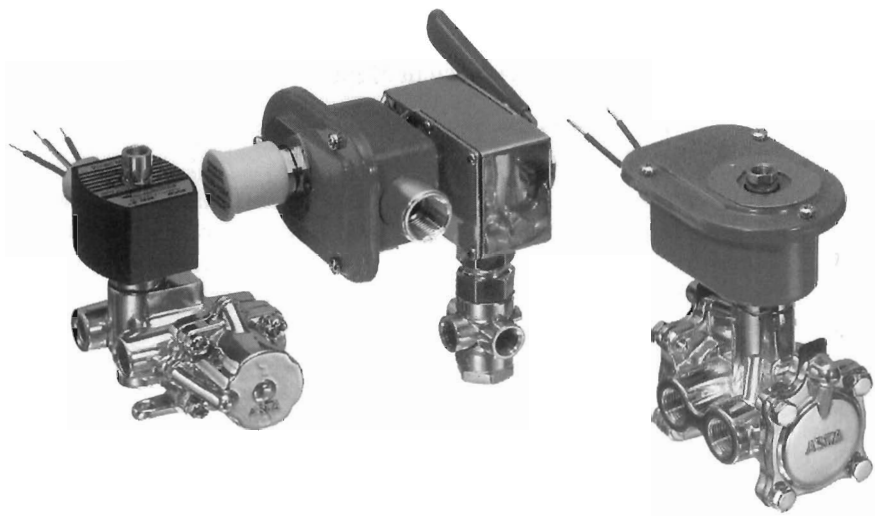
A solenoid valve is basically a valve operated by a built-in actuator in the form of an electrical coil (or solenoid) and a plunger. The valve is thus opened or closed by an electrical signal, being returned to its original position (usually by a spring) when the signal is removed. Solenoid valves are produced in two modes—normally-open or normally-closed (referring to the state when the solenoid is not energised).

The solenoid itself may be operated by d.c. or a.c. A d.c. supply may be provided by a battery, d.c. generator or through a rectifier. An a.c. supply is normally taken from a.c. mains voltage, through a transformer if necessary.

Valve types and classifications

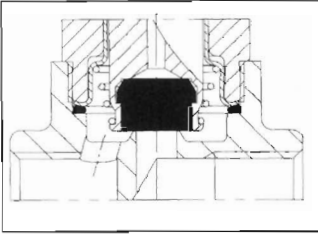
Basically there are nine different valve types to be distinguished, as shown in Figure 1.

Solenoid valves are also classified by type as follows:

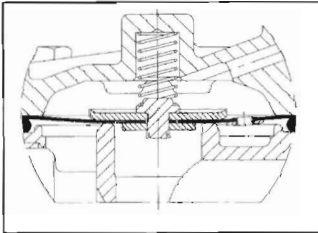


Solenoid valves.

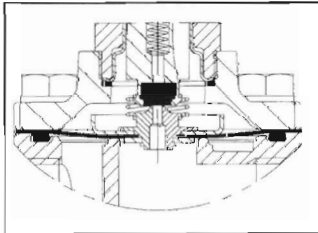
CORE DISC



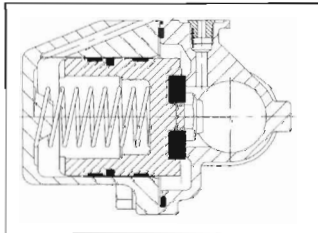
FLOATING DIAPHRAGM



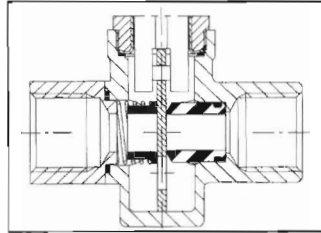
HUNG DIAPHRAGM



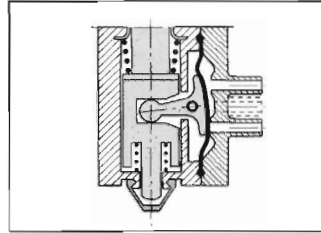
PISTON



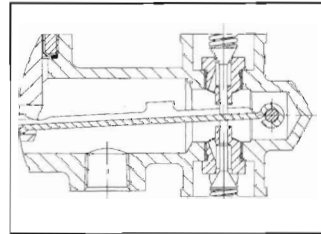
SLIDE DISC



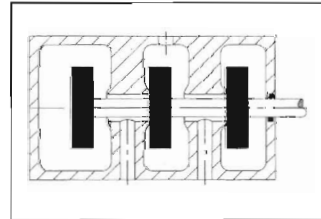
FLAPPER DISC



LEVER



POPPET



SPOOL

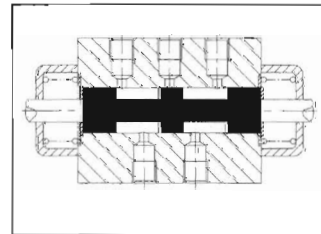


Figure 1. Solenoid valve types.

2/2 (2-way valves)

Two-way valves have one inlet and one outlet pipe connection (see Figure 2). They are of either

- normally-closed construction—valve is closed when de-energised and open when energised, or
- normally-open construction—valve is closed when energised and open when de-energised.

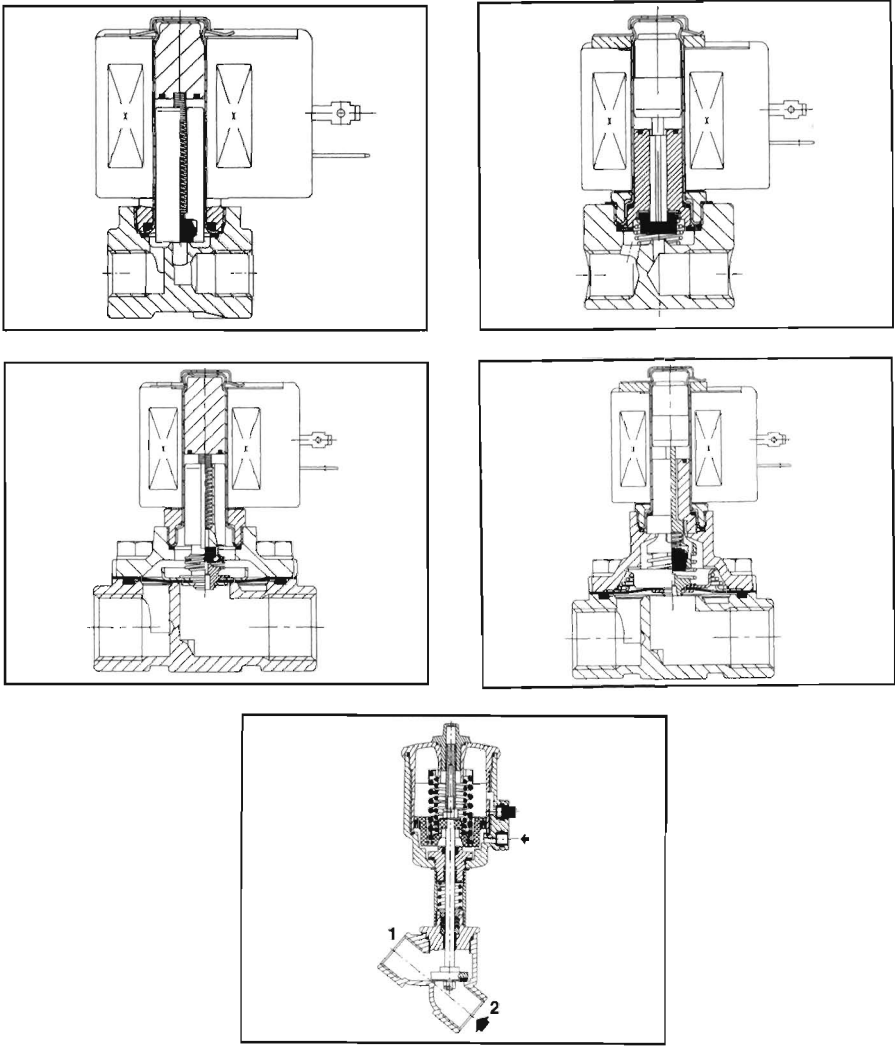


Figure 2. 2/2 (2-way valves).

3/2 (3-way valves)

Three-way solenoid valves have three pipe connections and two orifices (when one is open the other is closed and vice versa). They are commonly used alternately to apply pressure to and exhaust pressure from a diaphragm valve or single acting cylinder (see Figure 3). Three modes of operation are available.

- Normally-closed construction—with the valve de-energised, the pressure port is closed and the exhaust port is connected to the cylinder port. With the valve energised, the pressure port is connected to the cylinder port and the exhaust port is closed.
- Normally-open construction—when the valve is de-energised the pressure port is connected to the cylinder port and the exhaust port is closed. When the valve is energised, the pressure port is closed and the cylinder port is connected to the exhaust port.
- Universal construction—this allows the valve to be connected in either the normally-closed or normally-open position. In addition, the valve may be connected to select one or two ports (selection) or to divert flow from one port to another (diversion).

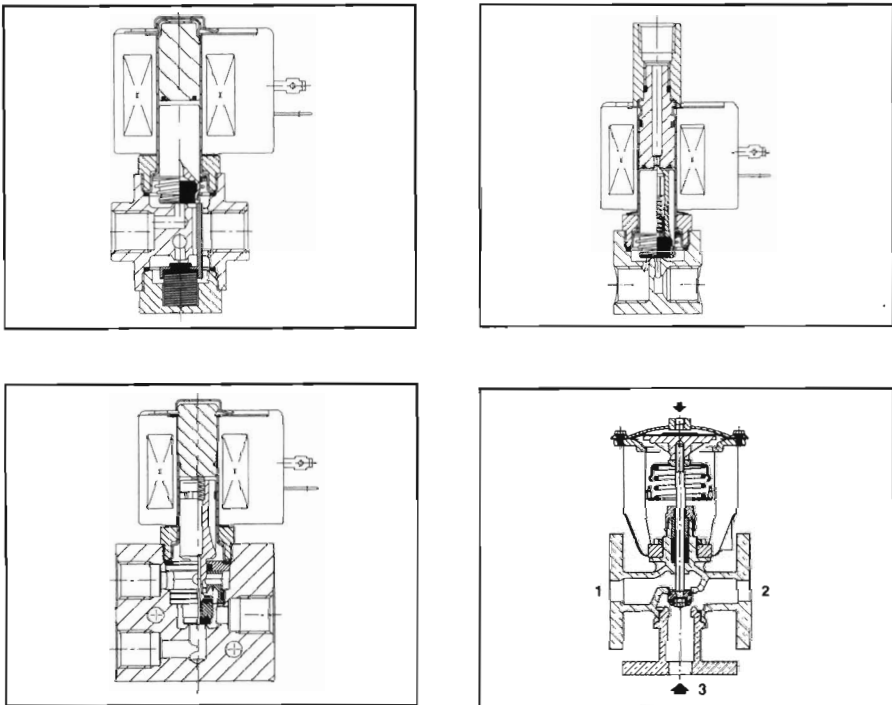


Figure 3. 3/2 (3-way valves).

4/2 and 5/2 (4-way valves)

Four-way solenoid valves are generally used to operate double-acting cylinders. These have four or five pipe connections—one pressure, two cylinder and one or two exhausts (Figure 4). In one valve position, pressure is connected to one cylinder port; the other is connected to the exhaust. In the other valve position, pressure and exhaust are reversed at the cylinder connections. Single-solenoid and dual-solenoid types are commonplace.

Manual reset valves

Manual reset valves must be manually latched into position and will return to their original position only when the solenoid has been energised or de-energised depending on construction (Figure 5). Four modes of operation are possible:

- electrically tripped—latched open
- electrically tripped—latched closed
- no voltage release—normally-closed
- no voltage release—normally-open

Valve classifications

- General purpose valve—a normally-open or normally-closed valve intended to control the flow of a fluid but not depended upon to act as a safety valve.
- Safety shut-off valve—a normally-closed valve of the 'on' or 'off' type, intended to be actuated by a safety control or emergency device to prevent unsafe fluid delivery. It may also be used as a general purpose valve. A multiple-port valve may be designated as a safety shut-off valve only with respect to its normally-closed port.

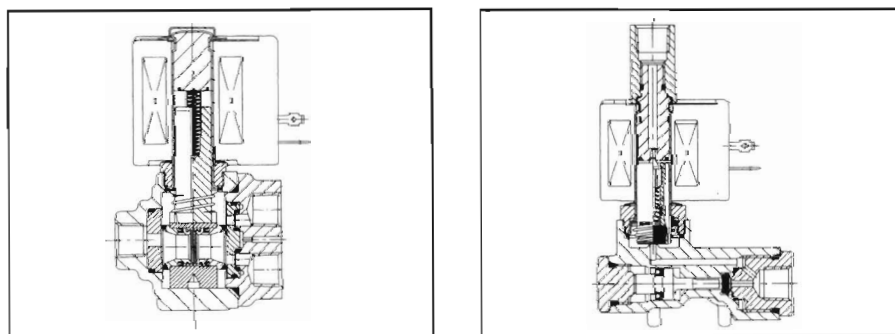


Figure 4. 4/2 and 5/2 (4-way valves).

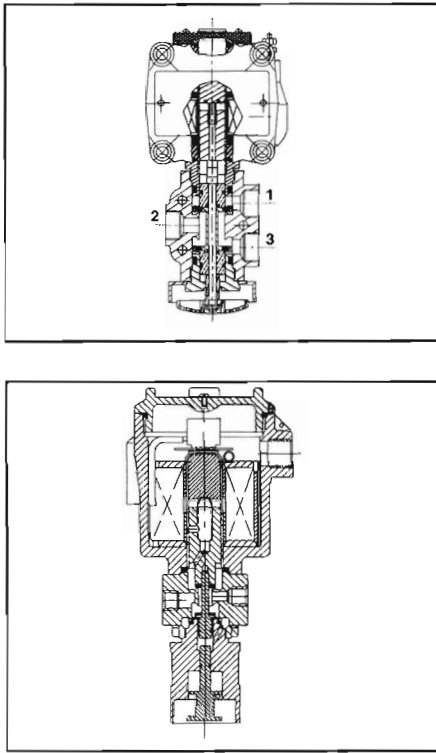


Figure 5. Manual-reset valves.

- Process-control valve—an approved valve (FM) to control flammable gases, but not to be relied upon as a safety shut-off valve.

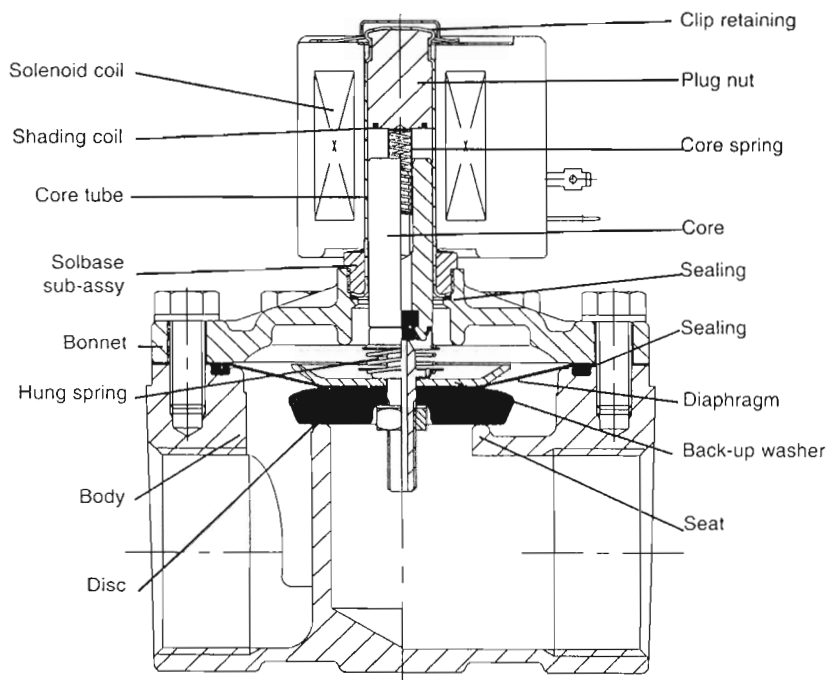
Direct-acting valves

When the solenoid is energised in a direct-acting valve, the core directly opens the orifice of a normally-closed valve or closes the orifice of a normally-open valve (Figures 6a and 6b).

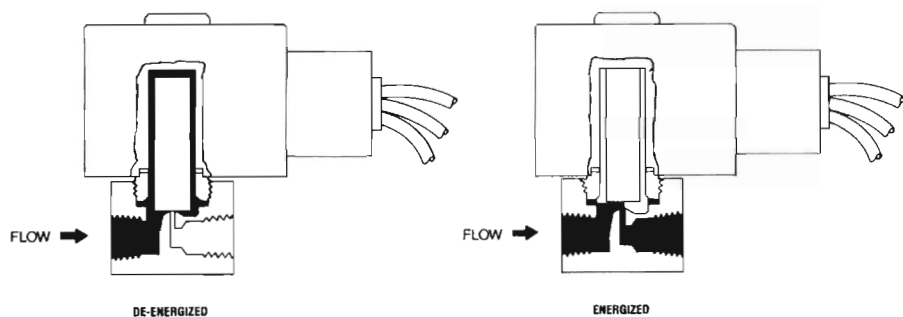
The force needed to open the valves is proportional to the orifice size and fluid pressure. As the orifice size increases, so does the force required. To open large orifices while keeping the solenoid size small, internal pilots are used.

Pilot-operated valve

This type of solenoid valve is equipped with a pilot and (smaller) (bleed) orifice and utilises the line pressure for operation. When the solenoid is energised, the pilot orifice is opened and releases pressure from the top of the valve piston or diaphragm to the outlet of the valve.



Direct-acting hung-diaphragm solenoid valve component parts.



Figures 6a and 6b. Direct-acting solenoid valves.

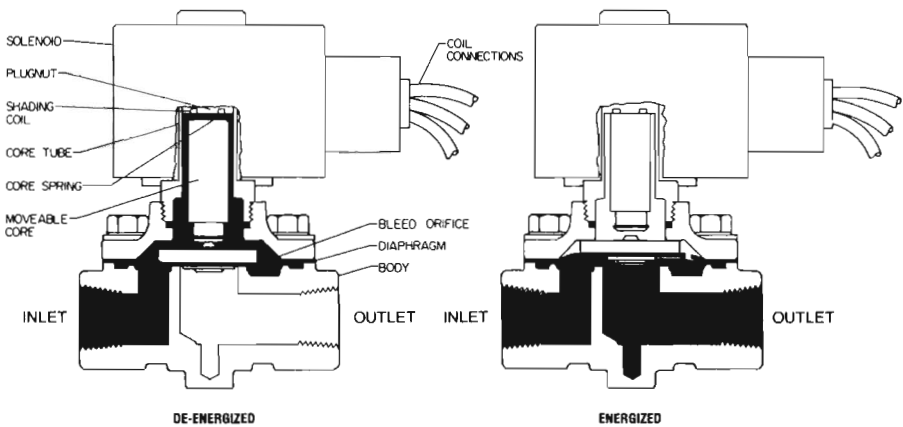
This results in an unbalanced pressure which forces the line pressure to lift the piston or diaphragm off the main orifice and open the valve.

When the solenoid is de-energised, the pilot orifice is closed and full line pressure is applied to the top of the piston or diaphragm through the bleed orifice, producing a sealing force for tight closure. There are two common types of construction:

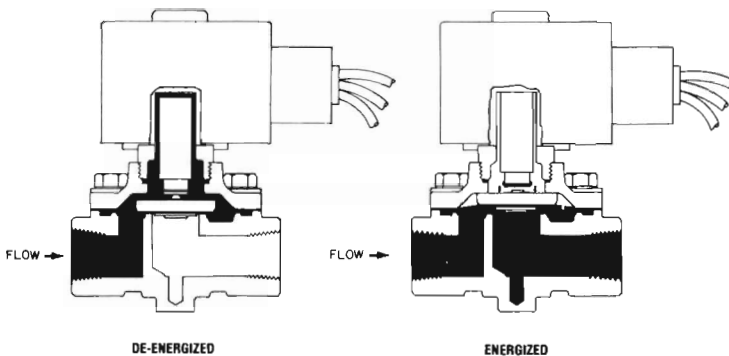
- (i) floating diaphragm or piston which requires a minimum pressure drop to remain in the open position (Figures 7a and 7b);
- (ii) hung-type diaphragm or piston which is mechanically held open by the solenoid core and operates from zero to the maximum pressure rating (Figures 8a and 8b).

Pressure-operated valve

This is a diaphragm- or piston-operated valve equipped with a 3- or 4-way solenoid pilot which alternatively applies a separate operating pressure to or from the diaphragm or piston to open or close the main valve (Figures 9 and 10).



Figures 7a and 7b. Internal pilot-operated floating-diaphragm valve.



Figures 8a and 8b. Internal pilot-operated hung-diaphragm valve.



Pressure-operated diaphragm solenoid valves.

Air-operated solenoid valves

An air-operated solenoid valve has two basic functional units:

- (i) an operator with a diaphragm or piston assembly which when pressurised develops a force to operates;
- (ii) a valve containing an orifice in which a disc or plug is positioned to stop or allow flow.

Low and instrument air-pressure range operators usually have a diaphragm for operation and for pneumatic operation the operator is generally a piston.

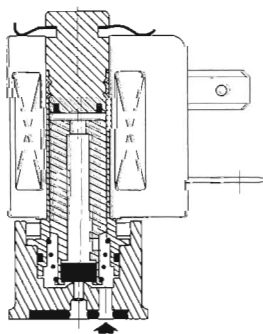


Figure 9. Pressure-operated diaphragm valve.

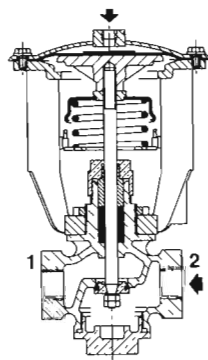
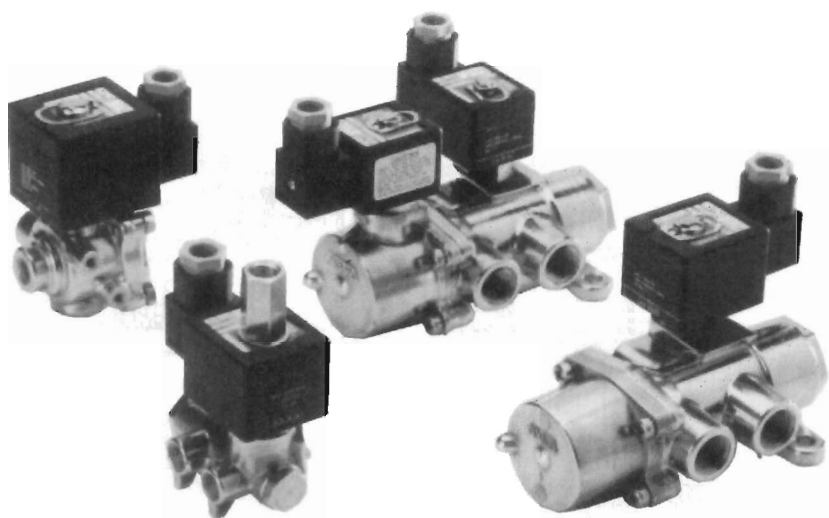


Figure 10. Pressure-operated piston valve.



4/2, 5/2 direct pilot-operated solenoid valves.

In dual-acting valves the operator stem is moved by the diaphragm or piston and directly opens or closes the orifice. In internal pilot-operated valves, the valve is equipped with a pilot and bleed orifice and utilises the line pressure for operation. Figure 11 shows a direct-acting valve flow diagram and Figure 12 an internal pilot-operated valve flow diagram.

Return-spring effect

With a two-way normally-closed valve, both the spring force and the fluid inlet pressure act to close the valve. As a consequence, the return spring can be made relatively weak, and in some designs eliminated entirely. The latter would require mounting the valve so that the solenoid was vertical, return action being by gravity plus fluid pressure.

With a two-way normally-open valve, the spring holds the valve open, assisted by fluid pressure. The solenoid force must be sufficient to overcome both spring pressure and inlet pressure to close the valve.

Three-way valves require an upper and a lower spring. The lower spring presses the valve against its seal opened by inlet pressure. The upper spring acts in a direction to force the valve open. The following are the combination of spring strengths required:

	Lower spring	Upper spring
Three-way normally-closed	Strong	Weak
Three-way normally-open	Weak	Strong
Mixer valve	Medium	Medium
Divider valve	Strong	Weak

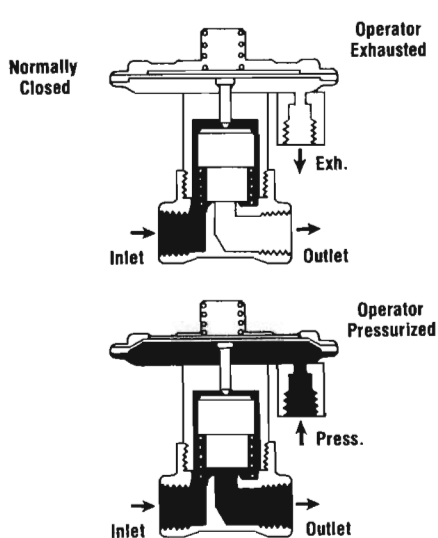


Figure 11. Flow diagram: direct-acting valve.

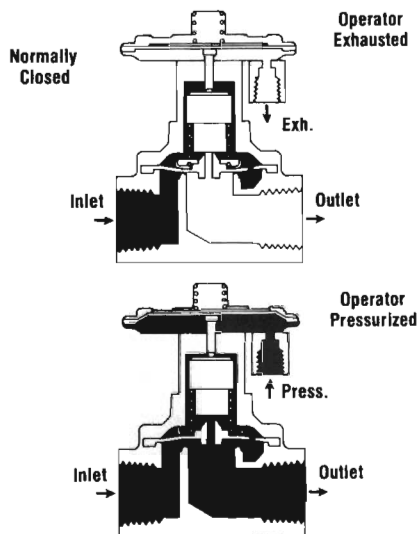


Figure 12. Flow diagram: pilot-operated valve.

Solenoid enclosures

Various types of enclosures may be used for solenoid coils, ranging from general purpose enclosures to protect from indirect splashing and dust, through dust and watertight enclosures to full explosion-proof enclosures. Requirements in this respect are specific to the application and selected accordingly.

Glandless solenoid valves

By arranging the solenoid armature to work in a sealed tube with the solenoid coil enveloping it, the sealing glands can be dispensed with, so simplifying the construction and eliminating one possible point of leakage.

This principle has been applied extensively to smaller valves. A typical type is shown in Figure 13.

This valve is T-shaped with two ports opposite each other, while the third is at right angles to them. The plunger, usually of a corrosion-resistant ferrous material, is spring-biased so that when unenergised it closes the lower orifice while leaving the other open. When energised, the plunger is pulled up so that the lower orifice is opened and the upper closed. If desired, the spring can be arranged to bias the plunger in one direction.

The plunger is provided with plastic valve discs, usually of synthetic rubber or nylon. Because the plunger is unbalanced, the force due to the pressure must be limited and the size of orifice, and therefore the flow and pressure drop, is usually related to the pressure.

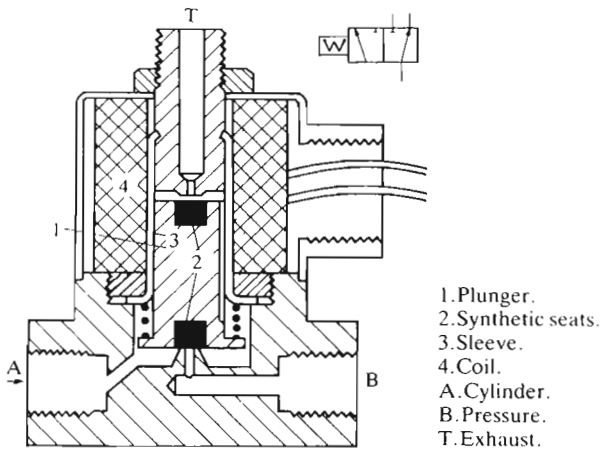


Figure 13. Glandless solenoid valve.

The maximum pressure is also related to the type of valve and may be as high as 210 bar (3000 lbf/in²) with an 0.85 mm ($\frac{1}{32}$ in) orifice. Flow depends on the allowable pressure drop and this in turn depends on the orifice size and the fluid.

Sealing is normally 'bubble-tight' but this is to some extent dependent on the cleanliness of the fluid. Lubrication is not essential but if used with air the valve life is increased by air-line lubrication.

Glandless valves can be installed in any position and will withstand appreciable shock loads. Response time is extremely short, 5 ms on a.c. and 10 to 15 ms on d.c. and it is said that speeds of up to several hundred cycles per minute are possible.

For hazardous atmospheres, most manufacturers supply explosion-proof materials which are slightly heavier and bulkier than the standard type.

Although these valves were originally developed for aircraft and missile application, there is no doubt that they have many uses in hydraulics, both as main valves for low-power systems and as pilot valves.

Where pressures do not exceed 17.5 bar (250 lbf/in²), a 1.6 mm ($\frac{1}{16}$ in) diameter orifice is suitable and this gives a flow of 2.31 l/min (0.50 gal/min) for 3.5 bar (50 lbf/in²) pressure drop. On a 50.8 mm (2 in) diameter cylinder, this would give a piston speed of 760 mm (30 in) per minute, while when used as a pilot valve for 25.4 mm (1 in) diameter spool with 12.7 mm (2 in) movement, the operating time is about half a second.

When acting as pilot valve, the actual flow would almost certainly be greater than that for a 3.5 bar (50 lbf/in²) pressure drop, as for a large part of the time the pressure drop will be nearer 14 bar (200 lbf/in²), until the resistance to piston or spool builds up.



Spool-type solenoid control valves.



Piston-type operator solenoid control valves.

Most of the manufacturers of these valves also supply pilot-operated valves incorporating the basic glandless valve; four-way valves, and two- and three-way valves for larger flows, are made in this way.

These valves may prove useful in acting as pilots to larger valves when the 'pressure release' principle is adopted. Figure 14 shows this applied to a differential-area spool valve. Normally the larger area keeps the spool to the right, with the solenoid valve SC closed. Opening the solenoid valve causes the pressure to drop in the left-hand chamber and the spring, with the excess pressure, causes the spool to move to the left. A two-position springless valve could also be operated in the same way, with jets and solenoid valves at each end.

Glandless solenoid valves—spool type

The construction of a four-way spring-centred closed-centre double-solenoid valve is shown in Figure 15. It has push solenoids with a spool movement of 1 mm (0.040 in) either side of the centre. It is suitable for pressures up to 140

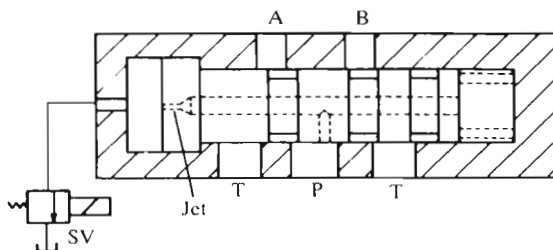


Figure 14. Small solenoid valve used as pilot, employing pressure-release principle.

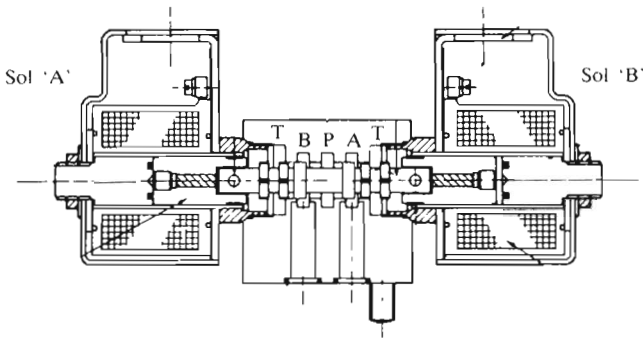


Figure 15. Glandless solenoid spool valve.

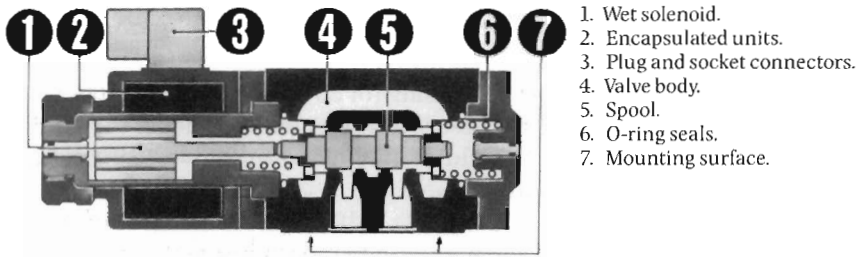


Figure 16. Wet-type solenoid valve.

bar (2000 lbf/in²) and has a flow of 9 l/min (2 gal/min) for a pressure drop of 2 bar (30 lbf/in²) on light hydraulic oil at 27 to 38°C (80 to 100°F). It is gasket-mounted and when used as a pilot valve is bolted on top of the main valve.

Another example of a glandless valve with a 'wet' armature is shown in Figure 16. All seals are static O-rings.

Selection

A number of operating and physical parameters must be considered when selecting a solenoid valve. The operating parameters include:

- Pressures

Maximum operating pressure differential (MOPD), i.e. the pressure the electrical solenoid has to overcome to open the valve and allow flow to occur.

Minimum operating-pressure differential (MinOPD), i.e. the minimum pressure drop that will exist across the valve when it is flowing.

Safe static pressure, i.e. the maximum pressure the valve can be subjected to in normal service.

Proof pressure, i.e. 5 times the safe working pressure.

- Temperatures

Normal ambient temperature

Maximum ambient temperature

Minimum ambient temperature

Maximum fluid temperature

- Viscosity

Viscosity is greatly dependent on temperature and to know the actual viscosity of a fluid, the real temperature of the fluid must be considered.

Oil grades—both hydraulic and fuel oils are classified relative to viscosity and are roughly distinguished in heavy and light oils.

- Response time

This is the time lapse after energising (or de-energising) a solenoid valve and depends on the valve size and operating mode, the kind of electrical supply, a.c. or d.c., fluids handled by the valve, temperature, inlet pressure and pressure drop.

Approximate values for a.c. valves on air service under average conditions are:

Small direct-acting valves: 5–10 ms

Large direct-acting valves: 20–40 ms

Internal pilot-operated valves:

(a) small diaphragm-type 15–60 ms

(b) large diaphragm-type 40–120 ms

(c) small piston-type 75–100 ms

(d) large piston-type 100–1000 ms

Operation on liquid media generally has little effect, i.e.

(a) small direct-acting valves: 20–30% higher

(b) large direct-acting valves: 50–150% higher

Response time on d.c. valves is approximately 50–60% higher than on a.c. operation.

- Valve seat tightness

Valve-seat tightness or leakage depends on the type of valve used, sealing materials, trim and medium.

Valve sizing

It is essential to size a valve properly because both undersizing and oversizing have undesirable effects.

Undersizing may result in:

- (1) Inability to pass desired flow requirements
- (2) Flashing of liquids to vapours on the outlet side of the valve
- (3) Lowering the outlet pressure
- (4) Creating a substantial pressure loss in a piping system

Oversizing may result in:

- (1) Unnecessary cost in oversized equipment
- (2) Variable flow through the valve or erratic control of the flow
- (3) Shorter life of some valve designs through oscillating of internal parts
- (4) Erratic operation of some designs such as failure to shift position due to lack of required flow in 3- and 4-way valves
- (5) Erosion of wire drawing of seats in some designs because they operate at nearly-closed position

The K_v (C_v) method of valve sizing reduces all variables to a common denominator called flow coefficient. Most, if not all, manufacturers provide extensive information and reference data for estimating K_v (C_v) and accurate sizing of solenoid valves. To summarise, the basic factors in valve sizing include:

- Maximum and minimum flows to be controlled
- Maximum and minimum pressure differentials across the valve
- Specific gravity, temperature and viscosity of fluids being controlled

Table 1 gives details of problems, possible causes and solutions in relation to the operation of solenoid valves.

It is recommended that manufacturers' data is referred to and followed carefully when problems occur. It is advisable that the manufacturer is consulted in cases of difficulty.

Table 1. Troubleshooting guide

Problem	Possible cause	Probable solution
Valve will not operate when valve circuit is energised (direct-acting valve)	Low voltage or no voltage to solenoid coil	Check voltage at coil: for most valves, voltage should be at least 85% of name plate rating.
	Burned out coil	See 'Coil failure' below.
	Excessive foreign matter jamming core in core tube	Clean valve; install strainer close to valve inlet.
	Binding core or damaged core tube	Replace parts.
	Excessive fluid pressure	Reduce pressure to valve nameplate pressure rating or install suitable valve.
Valve will not operate when valve circuit is energised (pilot-operated valve)	Same causes and solution as for direct-acting valve, plus:	
	Low pressure drop across valve	Valve might be oversized; replace valve with one having a smaller orifice. Increase pressure if possible.
	Ruptured diaphragm or piston ring	Replace damaged parts.
	Plugged or restricted pilot orifice	Clean valve and pilot orifice.
	Coil not de-energised	Check electrical control circuit.
Valve will not close or shift when valve circuit is de-energised (direct-acting valve)	Excessive foreign matter jamming core in core tube	Clean valve; install strainer close to valve inlet.
	Damaged disc or seat causing internal leakage	Replace with new parts.
	Binding core or damaged core tube	
	Damaged spring	Replace with new spring. Never elongate or shorten.
	Same causes and solution as for direct-acting valve, plus:	
Valve will not close or shift when valve circuit is de-energised (pilot-operated valve)	Plugged bleed orifice	Clean orifice.
	Damaged pilot seat or pilot disc	Replace with new parts.
	Damaged diaphragm or piston	Replace with new parts.

Damaged pilot spring	Replace with new spring. Never elongate or shorten spring.
Insufficient pressure drop across the valve	Valve might be oversized; replace valve with one having a smaller orifice. Increase pressure if possible.
Wire drawing	Replace valve body or install new valve; install suitable strainer close to inlet or valve.
Coil failure*	Check voltage at coil; voltage must conform to nameplate rating.
Overvoltage	Check for damaged core and core tube, or damaged spring. Check for scale or foreign matter on the core or inside the core tube. Clean thoroughly, and replace any damaged parts.
Damaged core or core tube causing inrush current to be drawn continuously	Check for damaged core and core tube, or damaged spring. Check for scale or foreign matter on the core or inside the core tube. Clean thoroughly, and replace any damaged parts.
Excessive foreign matter jamming core in core tube and causing inrush current to be drawn continuously	Reduce pressure or install suitable valve.
Excessive fluid pressure causing inrush current to be drawn continuously	Class A coils are limited to ambient temperatures of 77°F. For temperatures up to 167°F, use Class F coils; for temperatures up to 212°F, use Class H.
Excessive ambient or fluid temperature	Install missing solenoid housing and other metal parts or properly install incorrectly assembled metal parts. The housing and other metal parts form part of the magnetic circuit and are required to provide the impedance needed to limit current draw.
Missing solenoid parts	Waterproof the entrance conduit to prevent entry of moisture. If valve is mounted outdoors, check to see that enclosure is weatherproof and that gaskets are in good condition; use appropriate sealant where required. If general purpose enclosure is used in a damp or humid atmosphere, use watertight, moulded coils.
Moisture inside solenoid enclosure	

*In explosion-proof solenoids, a binding core, high-input voltage, or excessive ambient or fluid temperature may cause the solenoid's non-resettable thermal fuse to open. If this occurs, the solenoid must be replaced.